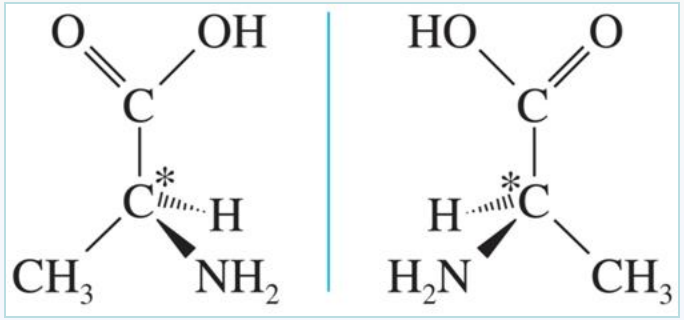
Chirality in Molecules Devoid of Chiral Centers

What is the necessary and sufficient condition for a molecule to be chiral?

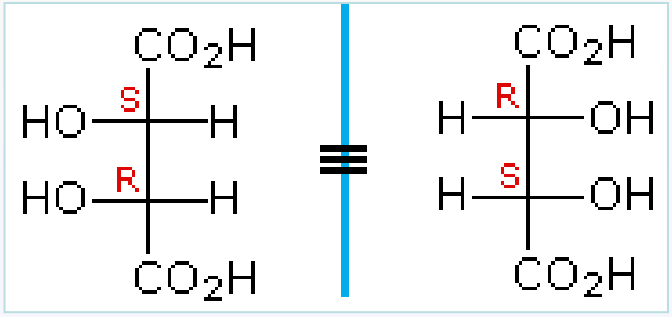
- A necessary and sufficient condition for a molecule to be chiral is that the molecule is non-superimposable with its mirror image.
- The presence of a (single, configurationally stable) chiral center in the molecule (central chirality) is a sufficient condition for the existence of chirality but not a necessary one.



One chiral center

Non-superimposable with its mirror image

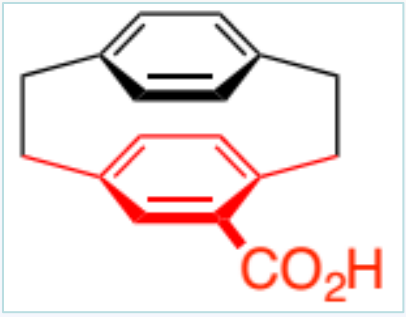
Optically Active



Two chiral centers

Superimposable with its mirror image

Optically Inactive

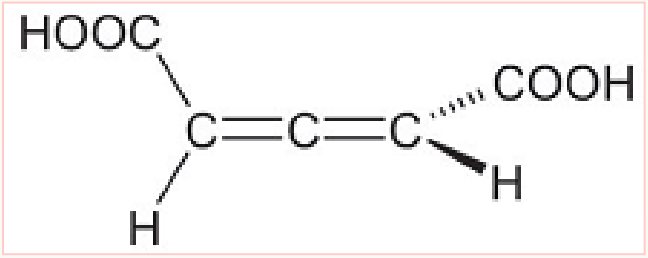


No chiral center

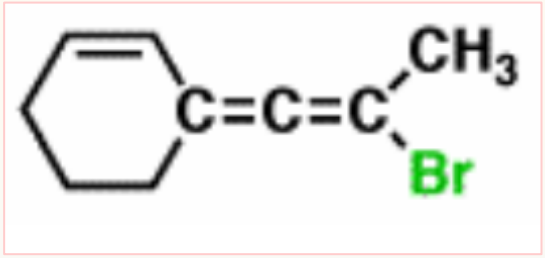
Non-superimposable with its mirror image

Optically Active

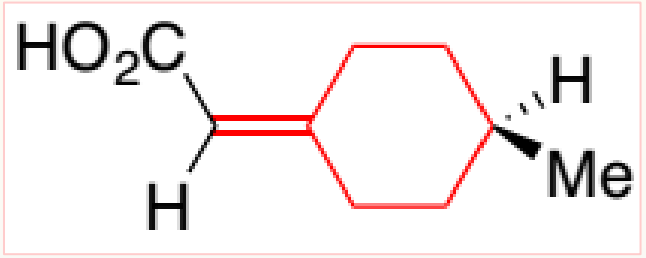
Chiral Molecules
Without Chiral Centers



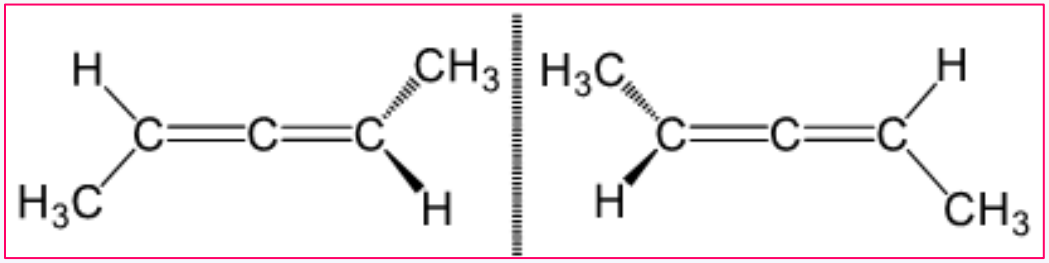
Allene



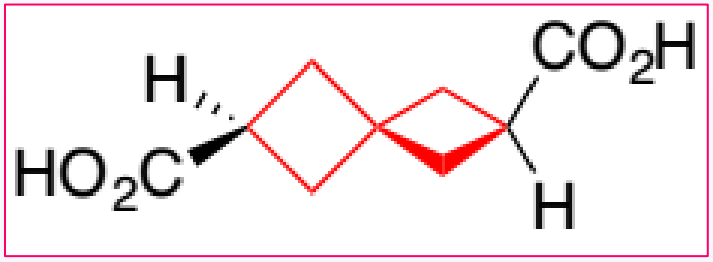
Allene



Alkylidenecyclohexane

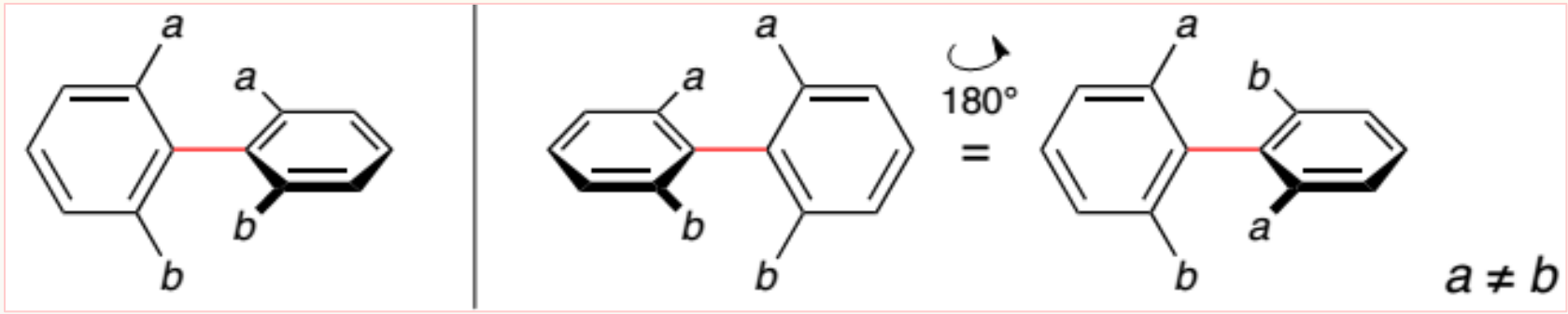


Chiral Allene
Non-superimposable with its mirror image

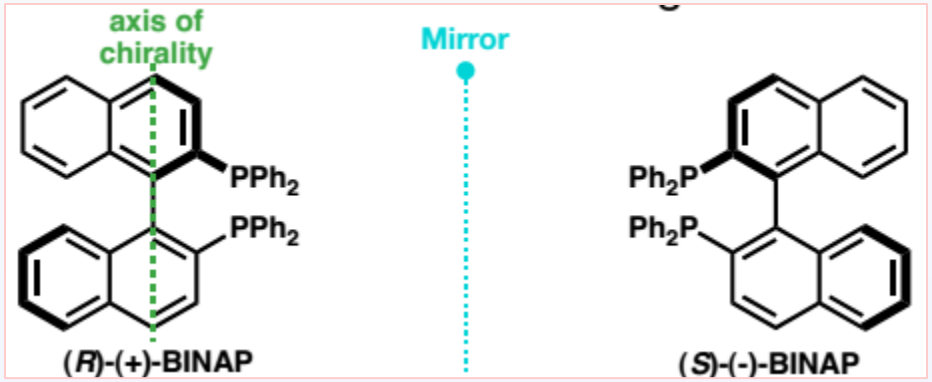


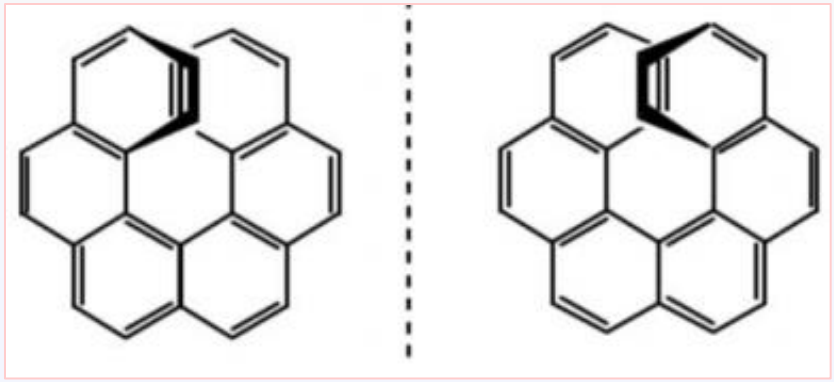
Spirane

Chiral Molecules Without Chiral Centers



Optically active biphenyl compound
Non-superimposable with its mirror image



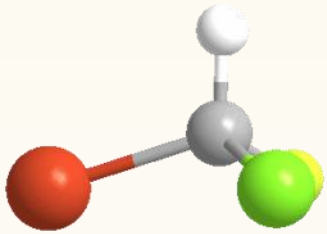


Chiral molecule with axis of chirality

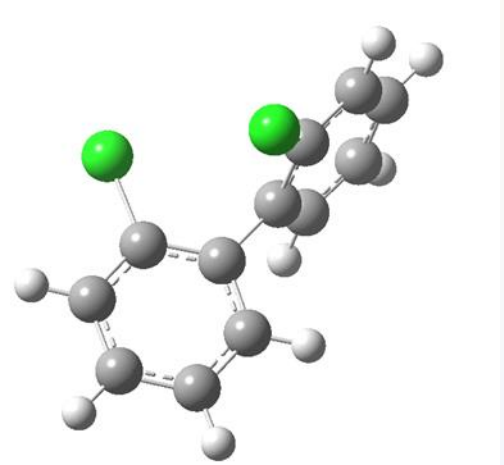
Chiral molecule with structural helicity

Elements of Chirality

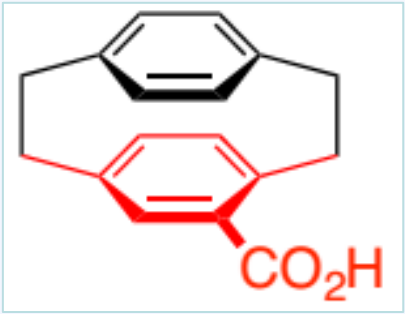
1. **Chiral Center**, or **Chirality Center**, or **Center of Chirality**



2. **Axis of Chirality, or Chirality Axis**



3. **Plane of Chirality, or Chirality Plane**

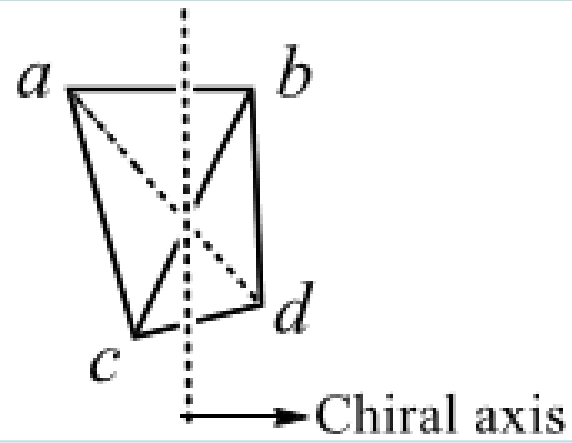


Molecules with Chiral Axes

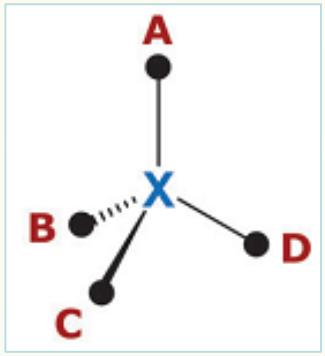
- Axial chirality is the chirality resulting from the nonplanar arrangement of four groups about an axis. This axis is called a chiral axis.

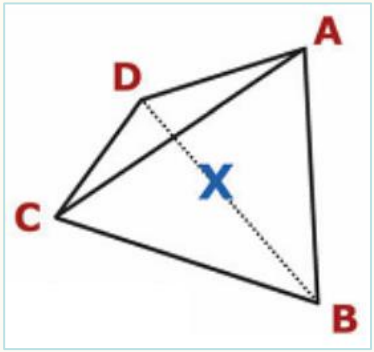
"Axial chirality is a special case of chirality in which a molecule does not possess a stereogenic center but an axis of chirality – an axis about which a set of substituents is held in a spatial arrangement that is not superimposable on its mirror image."

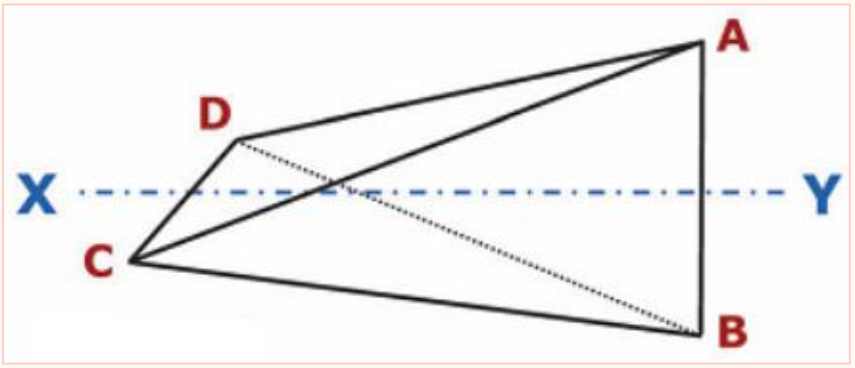
- With proper substitution, the long axis of this framework constitutes the chiral axis.
- A necessary and sufficient condition for chirality is that a ≠ b and c ≠ d.



Axial Chirality



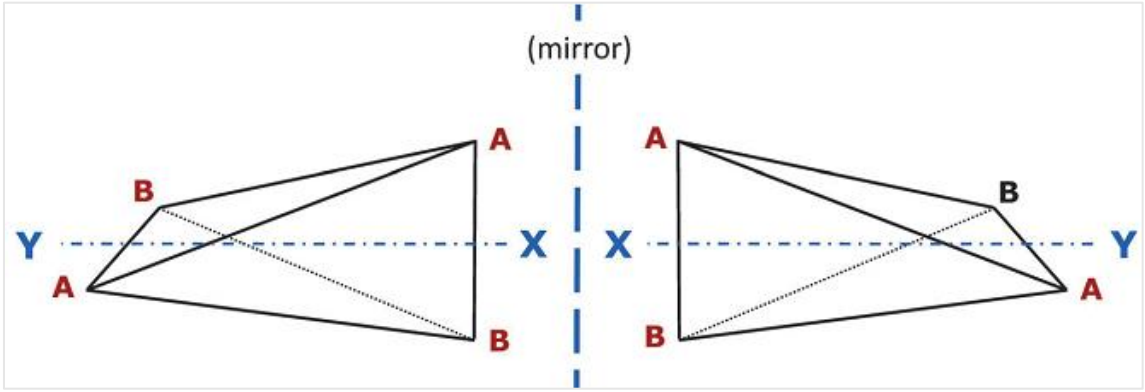




The regular tetrahedron symbolizes a central atom X with its four substituents A, B, C, and D located at the vertices. a stereogenic center exists when these four substituents are different.

If we now elongate the tetrahedron, the stereogenic center X is extended into an axis of chirality X–Y. the minimal condition for chirality here being simply A ≠ B and C ≠ D.

The two elongated tetrahedra are enantiomers despite having two A and two B substituents, the minimal condition for chirality here being simply A ≠ B.

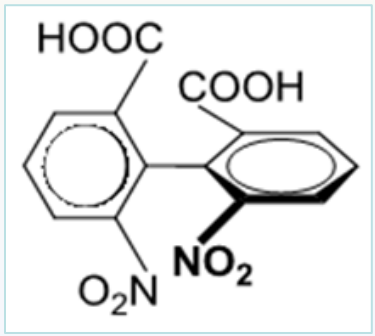


Atropisomerism



Richard Kuhn
(1900-1967)
Nobel **Laureate**
(1938)

- The term "Atropisomerism" was coined by Richard Kuhn in 1933.
- However "Optical activity due to axial chirality" was first reported by Christie and Kenner in 1922. Axial chirality was detected in 6,6'-dinitro-2,2'-diphenic acid.

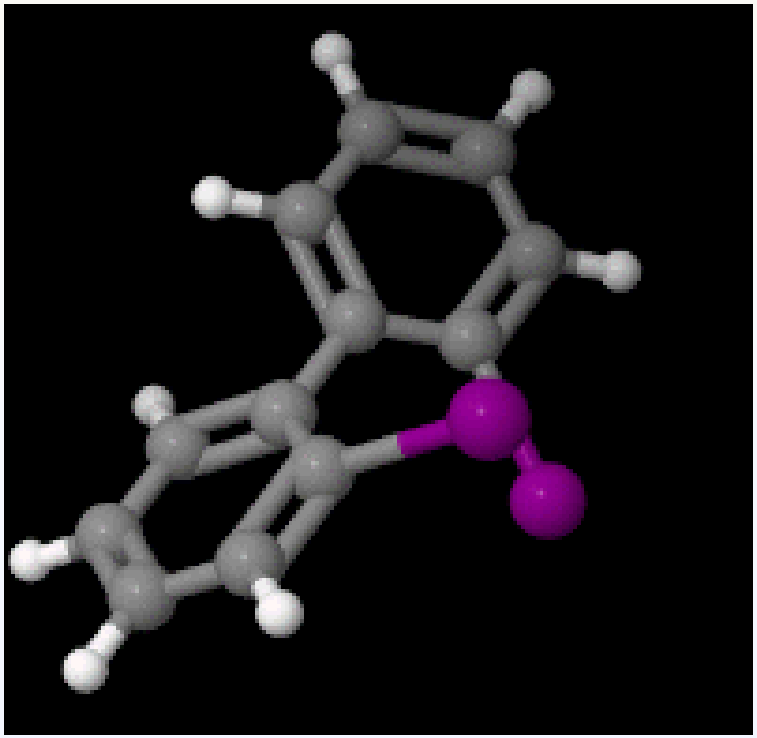


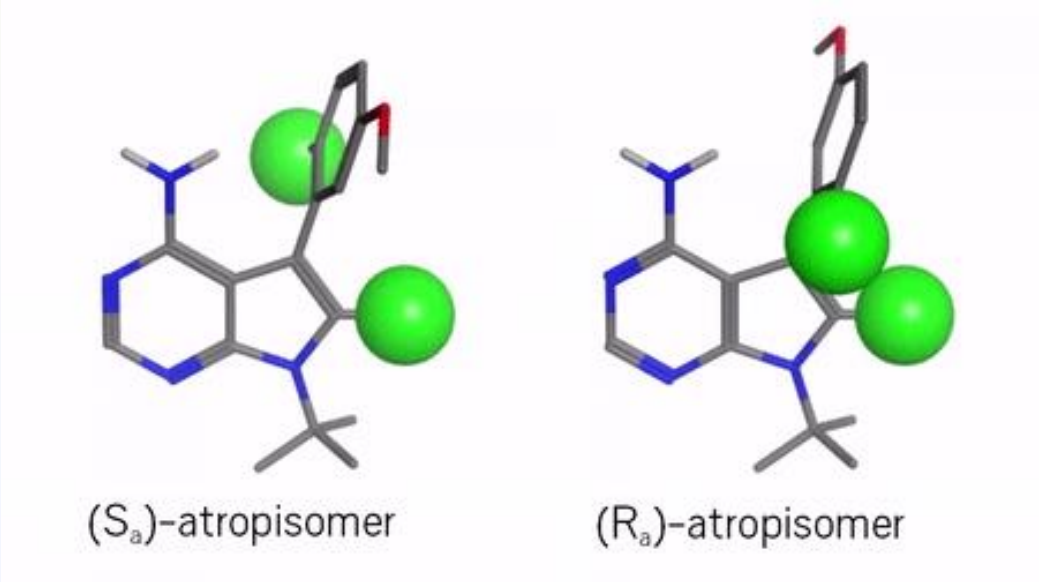
Single isomer was isolated from the racemic mixture *via* diastereoselective crystallization with a chiral resolving agent

- Optical isomers produced due to restricted rotation about a single bond is called atropisomers.
- Restricted rotation occurs when *o*-positions contain *two different bulky groups* and hence molecule is optically active.

Atropisomerism

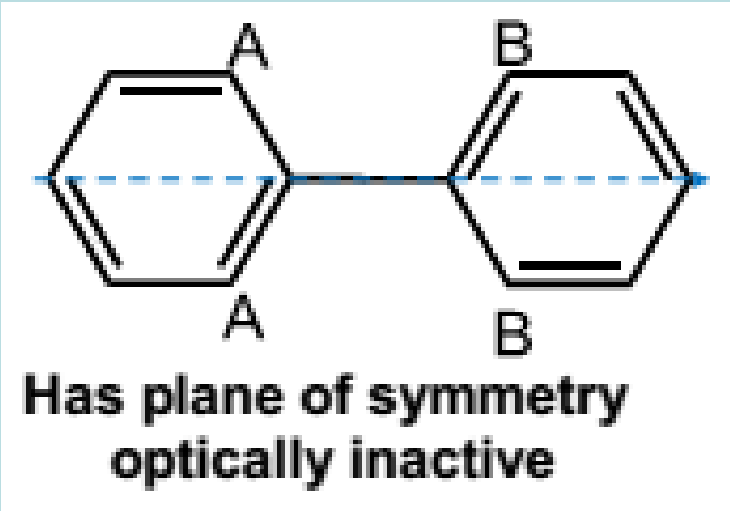
- **Atropisomerism** occurs when there is hindered rotation about a single bond as a result of steric or electronic constraints.



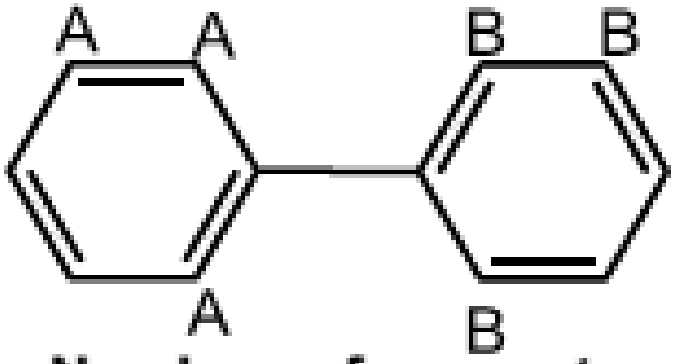


Optical Activity Of Biphenyl Compounds

- Conditions to exhibit optical activity for biphenyl compounds:
  - Neither ring must have a plane of symmetry.
  - Ortho-positions must be occupied by large groups or atoms.

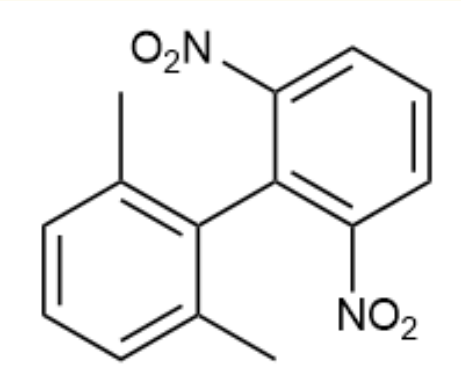


Has plane of symmetry
optically inactive

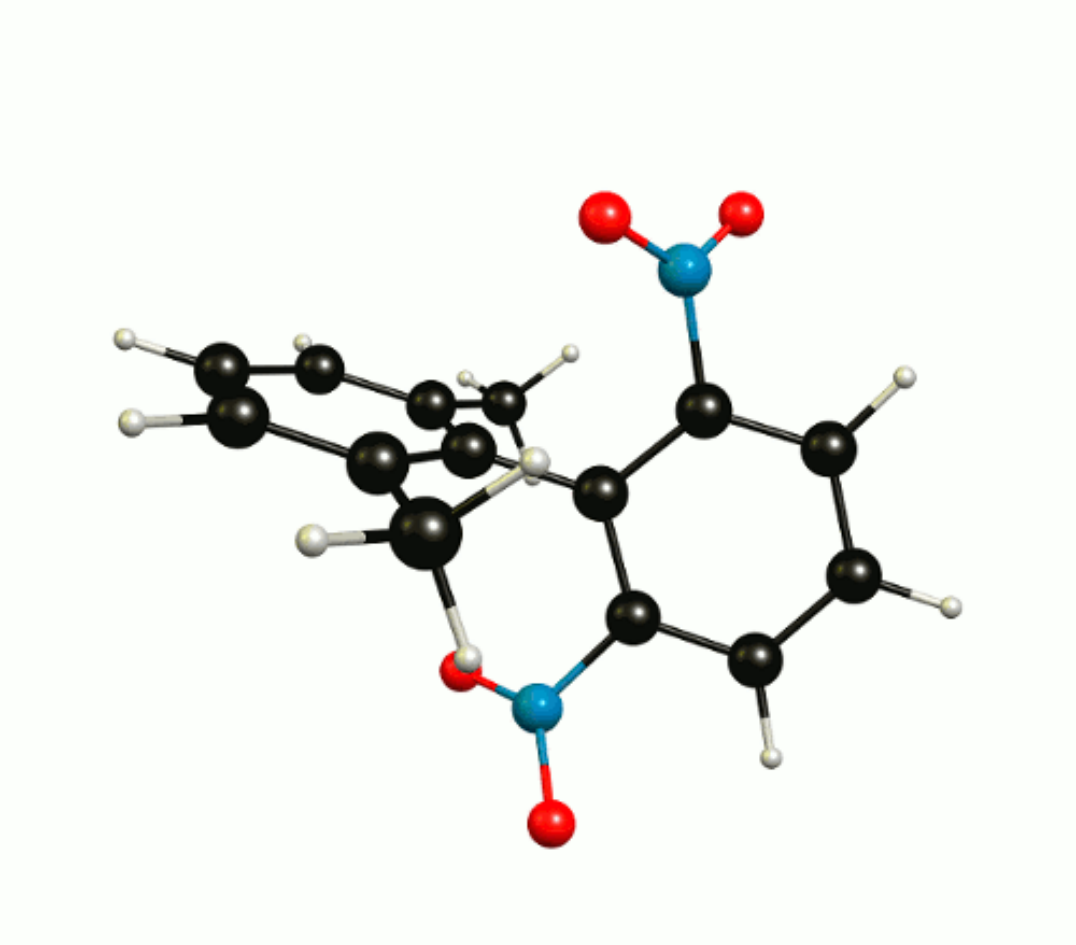


No plane of symmetry
optically active,

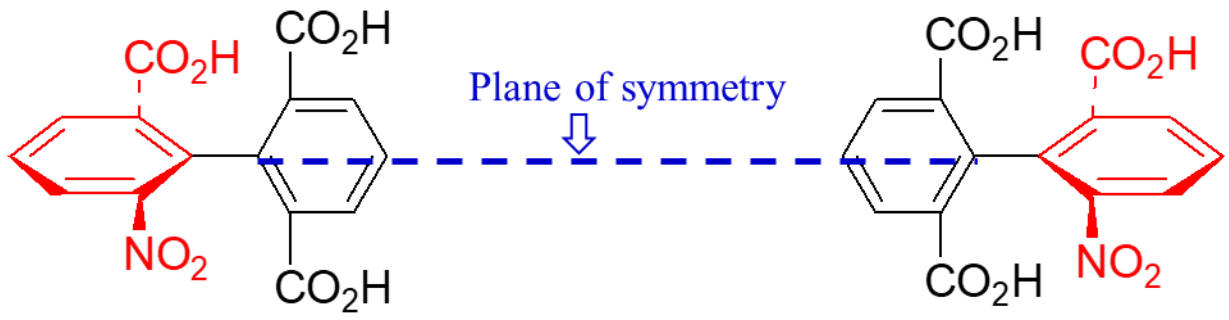
Optical Activity Of Biphenyl Compounds



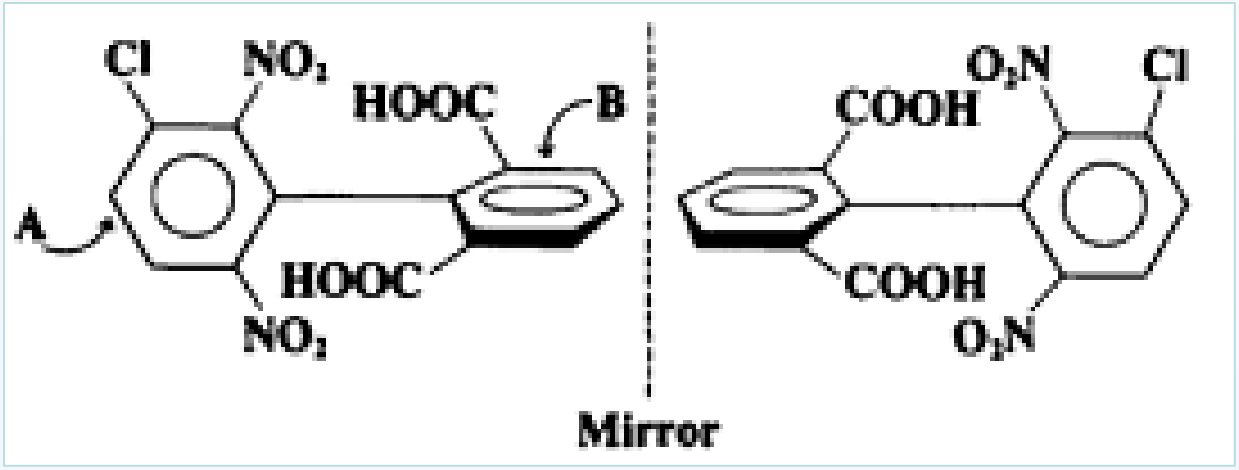
2,6-dimethyl-2',6'-dinitro-1,1'-biphenyl



Optical Activity Of Biphenyl Compounds



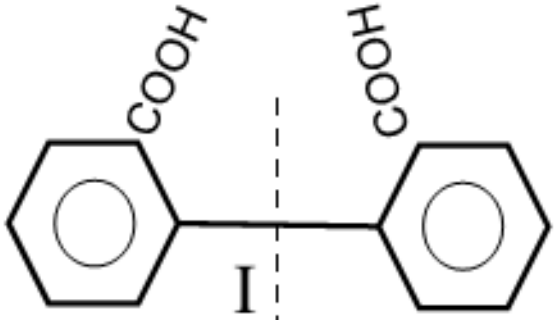
- Optically inactive due to presence of plane of symmetry



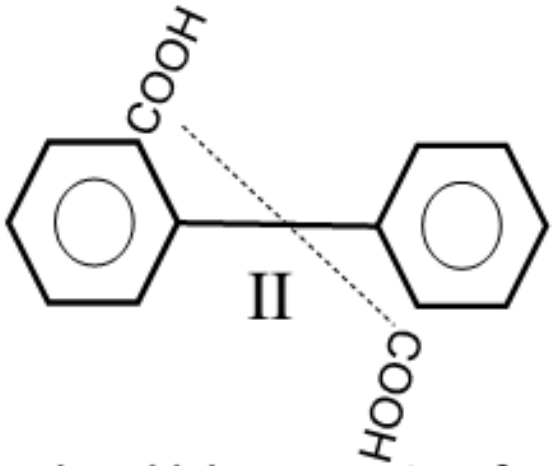
- Ring B is symmetrically substituted. A plane of symmetry exists.

The compound is achiral

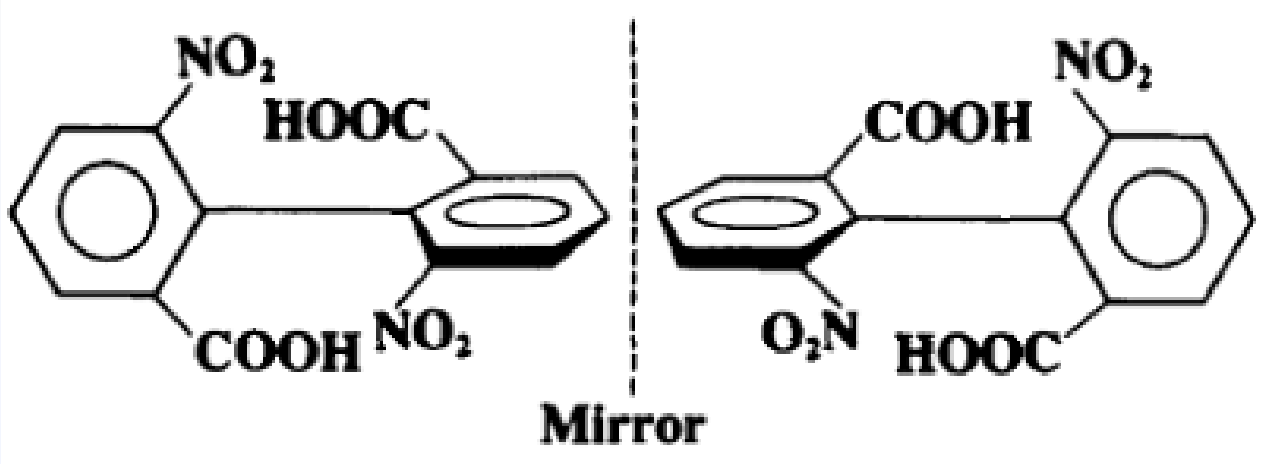
Optical Activity Of Biphenyl Compounds



Diphenic acid, has a plane of symmetry
Optically inactive



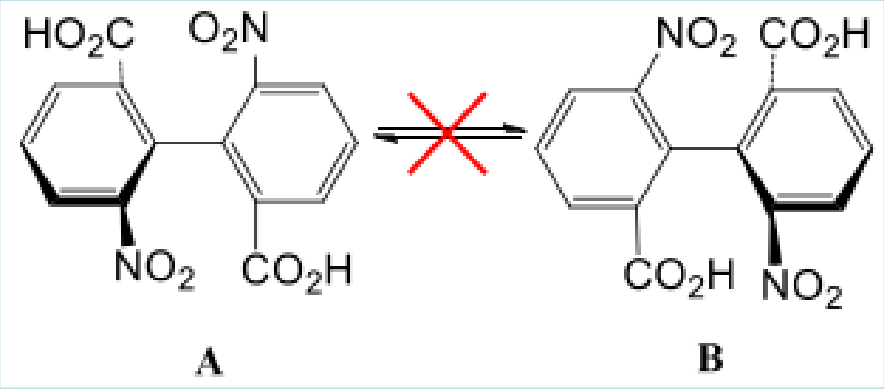
Diphenic acid, has a centre of symmetry
Optically inactive

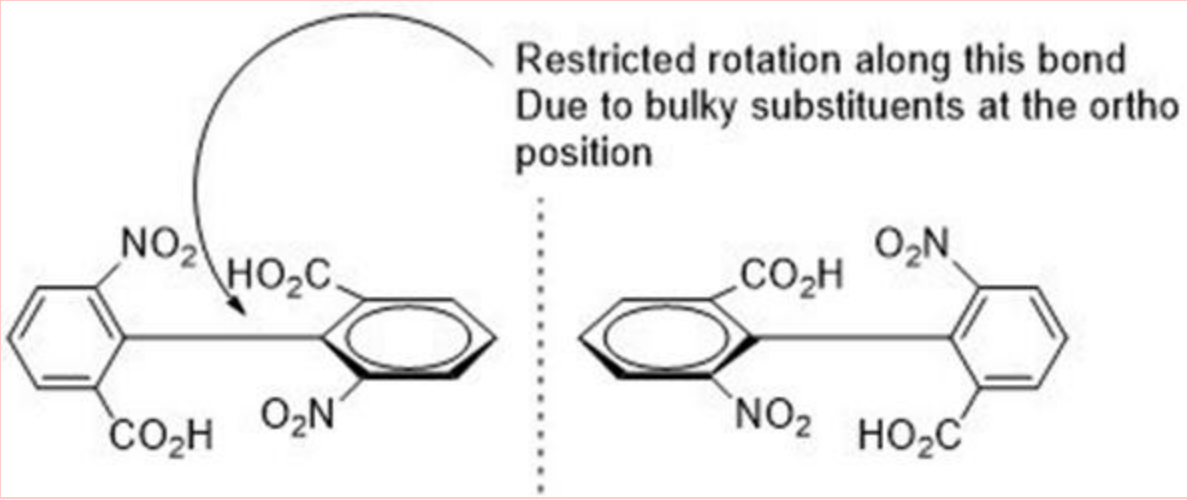


- No plane of symmetry, so optically active.

Optical Activity Of Biphenyl Compounds

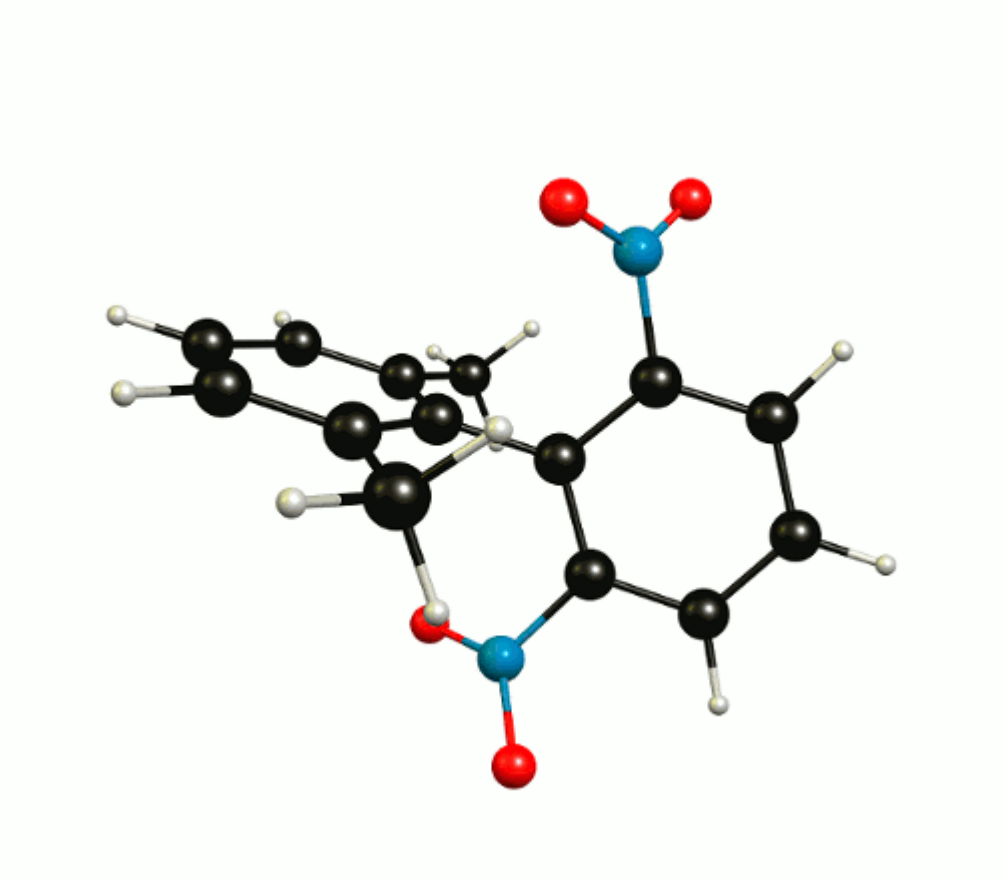
- Conformers that cannot interconvert (due to steric hindrance) can be enantiomers.





Non-coplanar, optically active

Chirality in Molecules Devoid of Chiral Centers



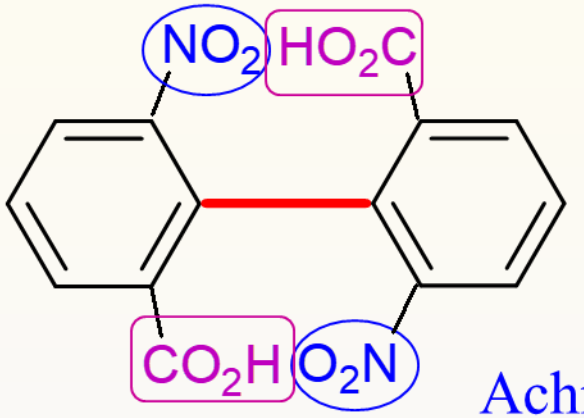
Atropisomers

- Atropisomers are stereoisomers resulting from restricted rotation about a single bond where the steric strain barrier to rotation is high enough to allow for the isolation of the conformers.

- Large groups had to be present in the *ortho*-positions to behave as "obstacles" to prevent free rotation about the single bond joining two benzene rings. Thus the two benzene rings can not be coplanar, rather they (benzene rings) become perpendicular to each other, and hence the molecule becomes chiral. Thus optically active biphenyl compounds can be isolated due to restricted rotation $\Rightarrow$ **obstacle theory**.

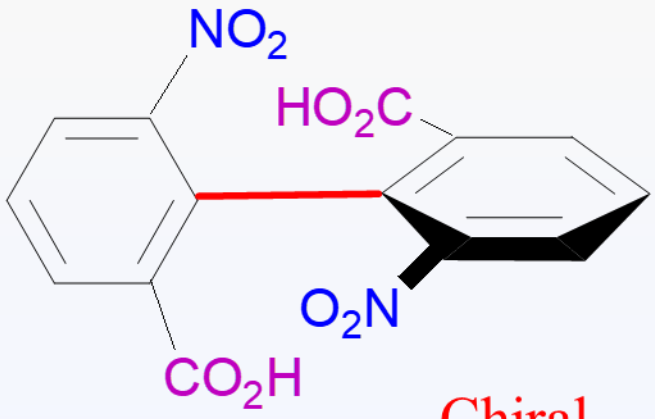
Atropisomers

Impossible, too crowded

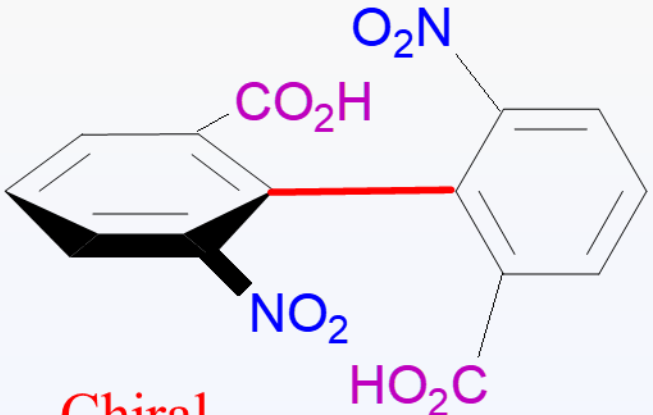


Achiral

Due to the presence of large *ortho*-substituents, two benzene rings can not be coplanar



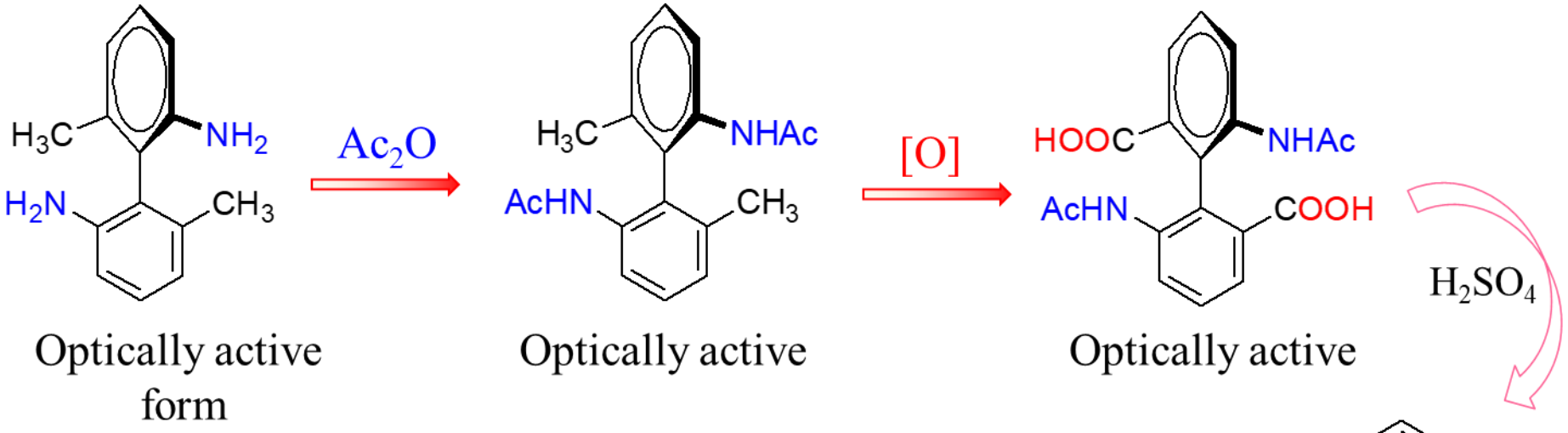
Chiral



Chiral

- In optically active biphenyl compounds, two rings tend to be perpendicular.

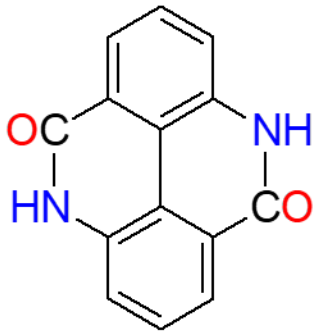
Evidence for Obstacle Theory



Optically active form

Optically active

Optically active



Center of symmetry present

Optically inactive

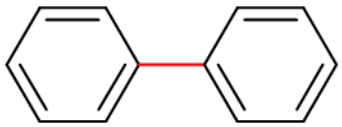
- Evidence for the obstacle theory, *i.e.*, interference of groups, amounts to proving that the two rings in optically active compounds are not coplanar.
- In all the optically active compounds, the rings can not be coplanar.
- The dilactam is planar and has a center of symmetry, so it is optically inactive

Evidence for Obstacle Theory

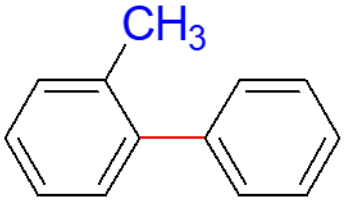
- Ultraviolet spectra measurements of biphenyl compounds have shown that two rings in *o*-substituted biphenyls are not coplanar.



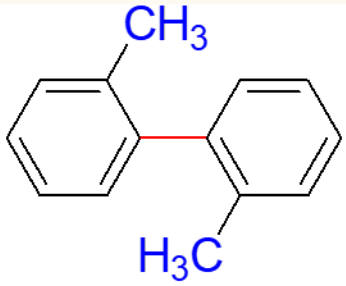
$\lambda_{max} = 198$ nm
(ε 8000)



$\lambda_{max} = 248$ nm
(ε 19000)



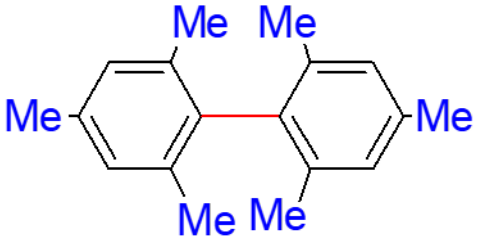
$\lambda_{max} = 236$ nm
(ε 10000)



$\lambda_{max} = 224$ nm
(ε ~ 700)

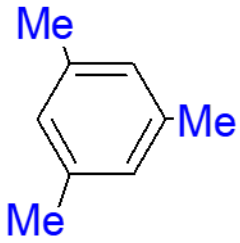
- When *o*-positions are occupied, the coplanarity is prevented, the maximum wavelength will shit to shorter wavelength.

- UV spectra of bimesityl and mesitylene are almost identical. In (I) coplanarity is prevented due to steric inhibition of resonance.



(I)

$\lambda_{max} = 267$ nm (ε 545)

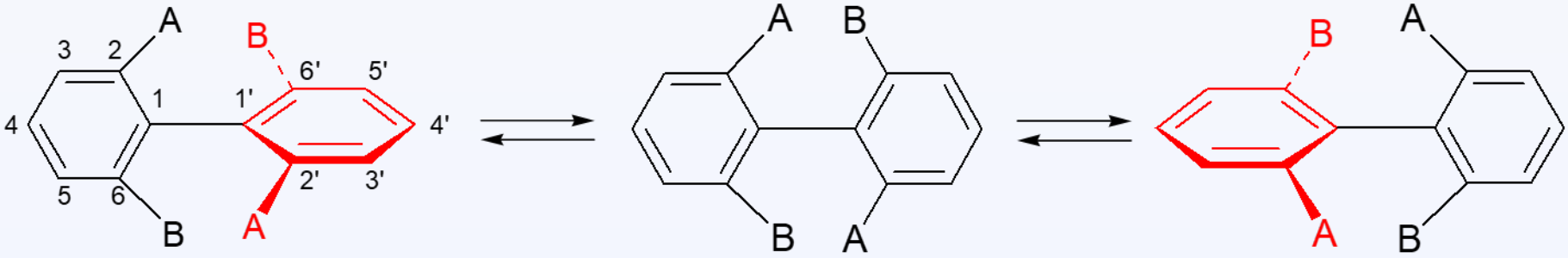


(II)

$\lambda_{max} = 266$ nm (ε 260)

Racemization of Biphenyl Compounds

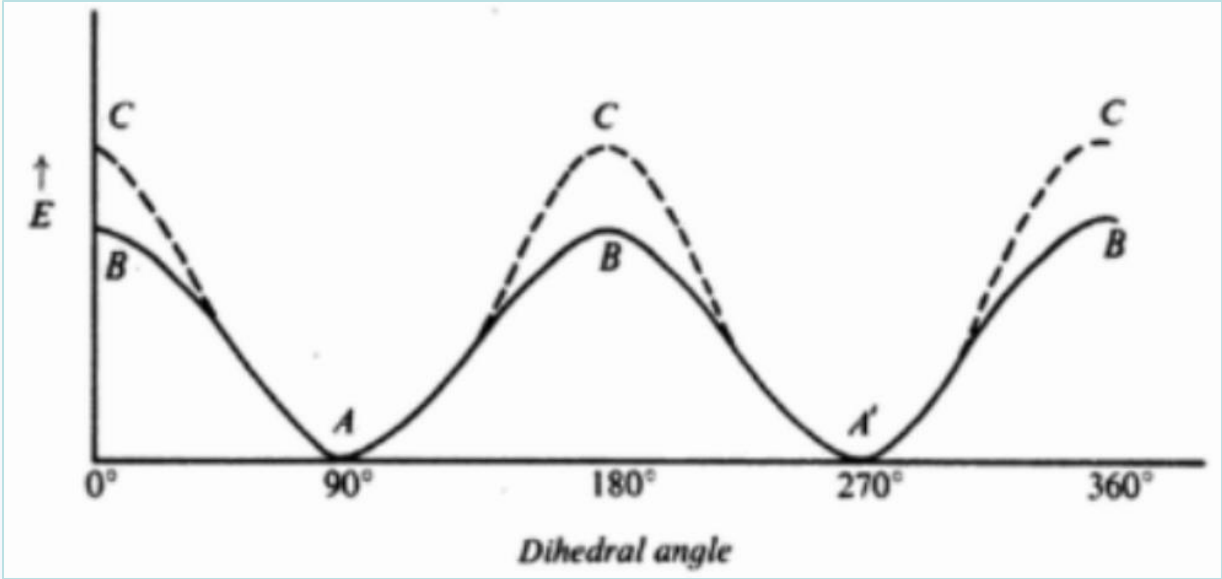
- Since the optical activity of biphenyl compounds arises from restricted rotation, it might be expected that racemization of these compounds would not be possible.
- In practice, many optically active biphenyl compounds can be racemized under suitable conditions, *e.g.*, boiling in solution.
- Heating increases the amplitude of the vibrations of the substituent groups in the 2,2′,6,6′-positions, and also the amplitude of vibration of the two benzene rings with respect to each other, thereby permitting the substituent groups to slip by one another.
- Thus the nuclei pass through a common plane. So the final product will contain equimolecular amount of the (+)- and (–)-forms.

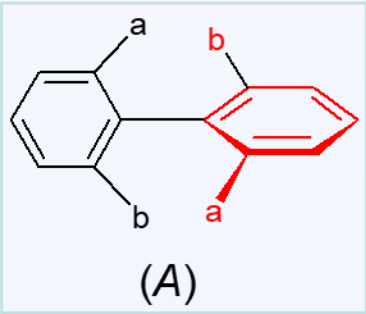


Planar form

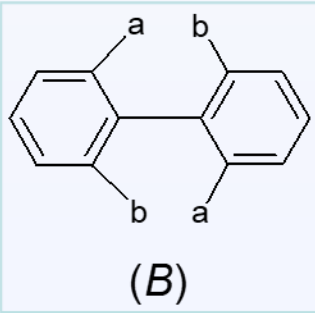
Racemization of Biphenyl Compounds

- The enantiomers (*A*) and (*A′*) have the same energy content. As they approach the planar configuration (B) the energy content increases due to steric repulsion.
- The larger the groups a and b, the higher will be the energy barrier, *i.e.*, the slower will be the rate of racemization.

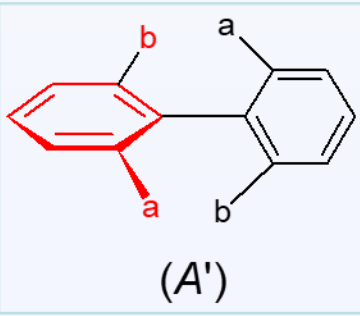




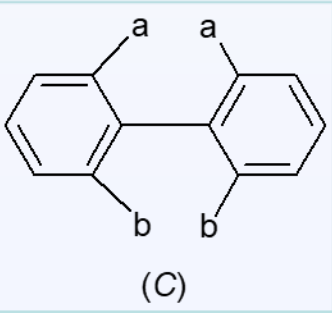
(*A*)



(*B*)



(*A'*)



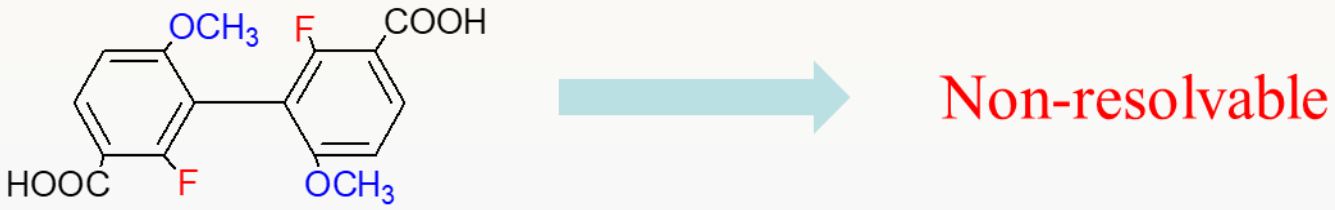
(*C*)

Racemization of Biphenyl Compounds

- 2,2′,6,6′-Tetrasubstituted biphenyl compounds may be classified under three headings according to the nature of the substituent groups:

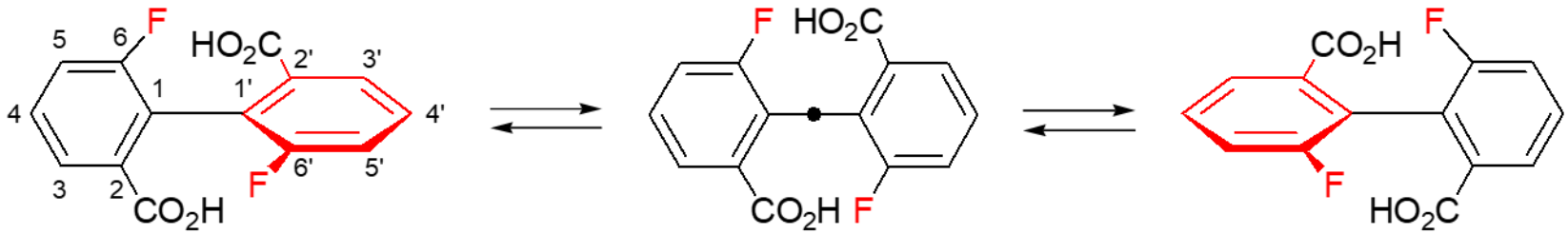
i. Non-resolvable: These contain any of the groups (H, -OCH_3 or F).

The effective volumes of these groups are too small to prevent rotation about the single bond.



ii. Resolvable but Easily Racemized: These must contain at least two –NH_2 or –COOH groups, or one –NH_2 and one –COOH group.

The remaining groups may be any of -OCH_3 or F , but not H.



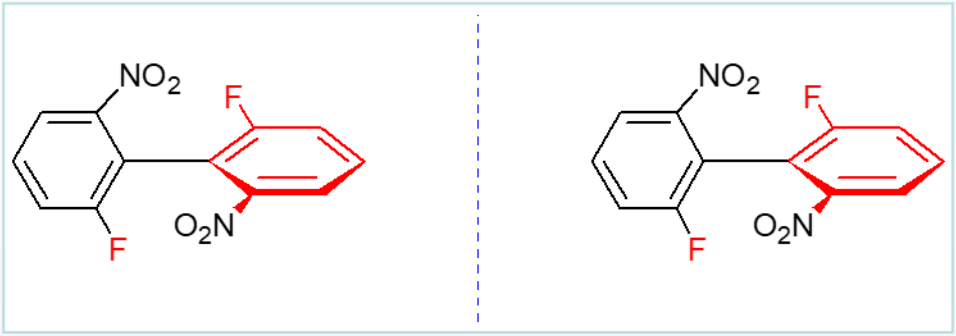
Planar form has a center of symmetry

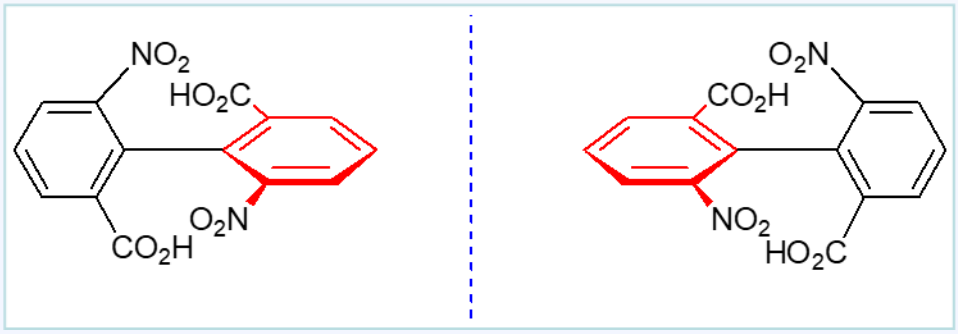
Racemization of Biphenyl Compounds

- 2,2′,6,6′-Tetrasubstituted biphenyl compounds may be classified under three headings according to the nature of the substituent groups:

iii. Not Racemizable at All: These biphenyl compounds contain least two $-NO_2$ groups.

The remaining groups may be any of $-OCH_3$, F, $-NH_2$ or –COOH groups but not H.

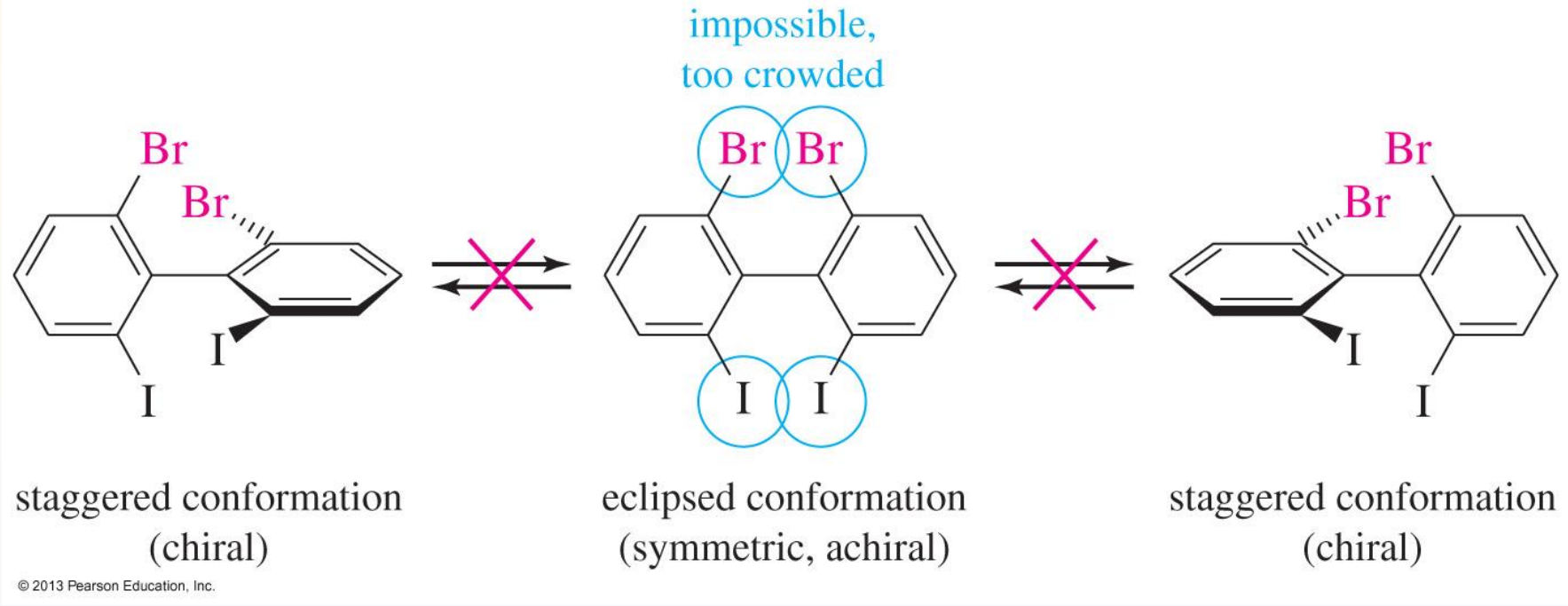




Resolvable and cannot be racemized.

Optically Active

Optical Activity of Biphenyl Compounds



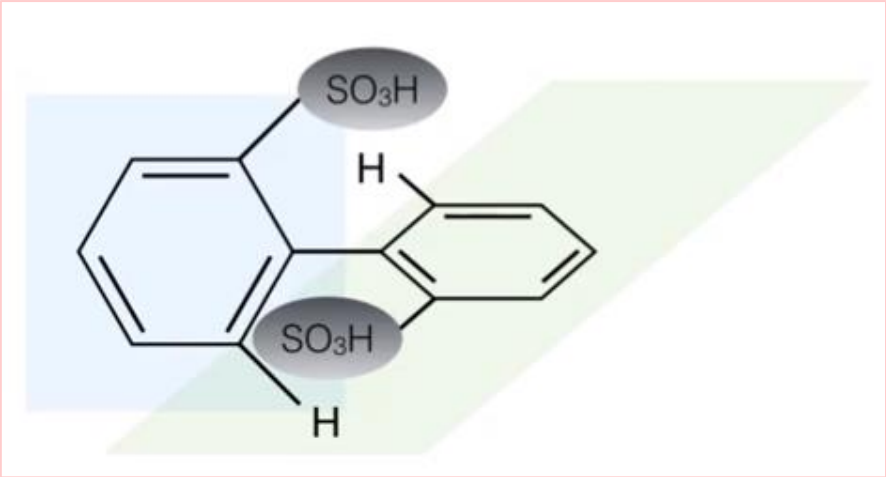
The order of steric hindrance produced by various groups:

$$Br >> Me > Cl > NO_2 > CO_2H >> OMe > F$$

- This order roughly parallels van der Walls radii.

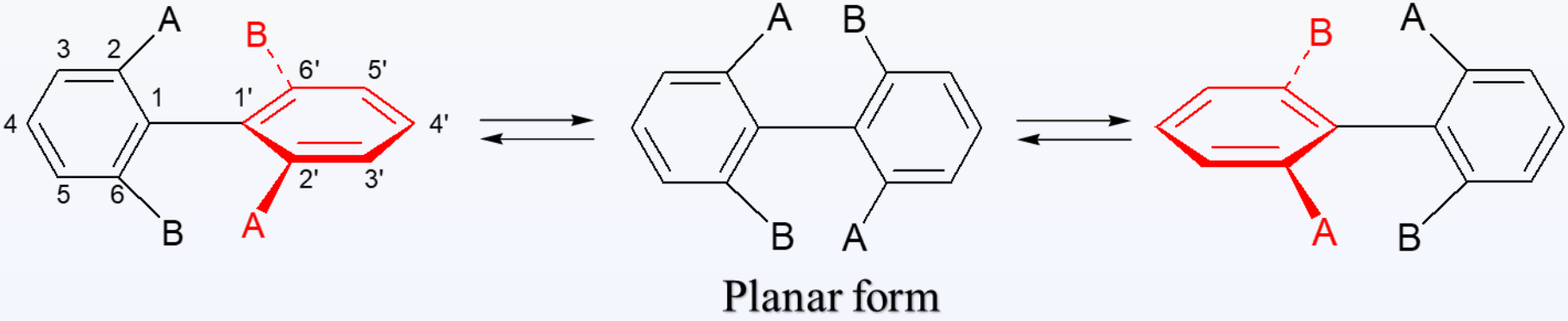
Optical Activity of Biphenyl Compounds

- It is not always necessary for four large *ortho* groups to be present in order for rotation to be prevented.
- Compounds with three and even two groups, if large enough, can have hindered rotation and, if suitably substituted, can be resolved.



- Diphenyl 2,2′-disulphonic acid, optically active, noncoplanar due to steric hindrance, readily recemized on heating.

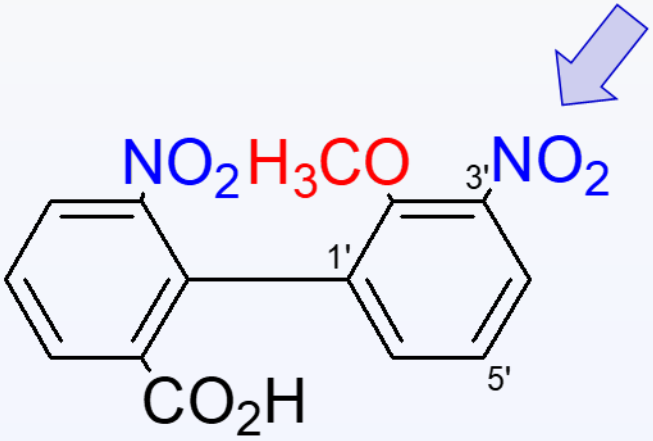
Chirality in Molecules Devoid of Chiral Centers



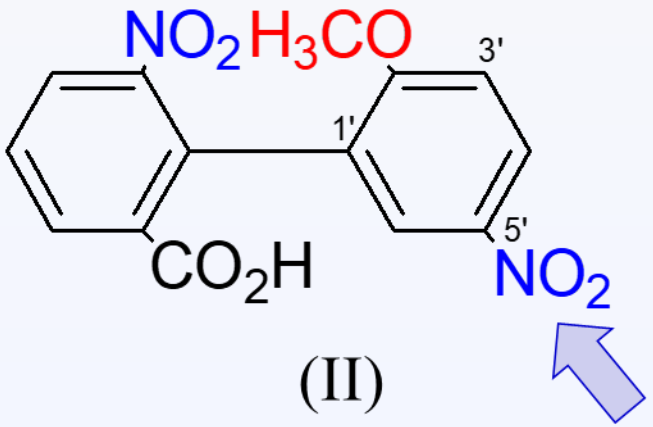
Planar form

Buttressing Effect

- In addition to the size of the *ortho*-substituents, the nature and position of other substituent groups also play a part in the rate of racemization.
- Substituents in the meta position tend to enhance racemization barriers by what is known as "buttressing effect".
- The rate of racemization of (I) is much slower than that of (II).
- The nitro-group in position 3′ has a much greater stabilizing influence than in position 5′.



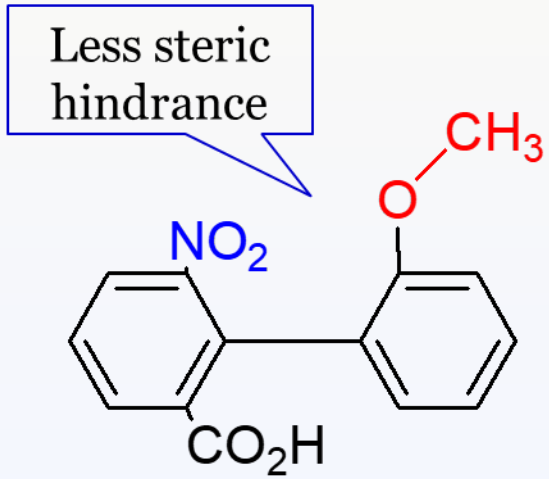
(I)



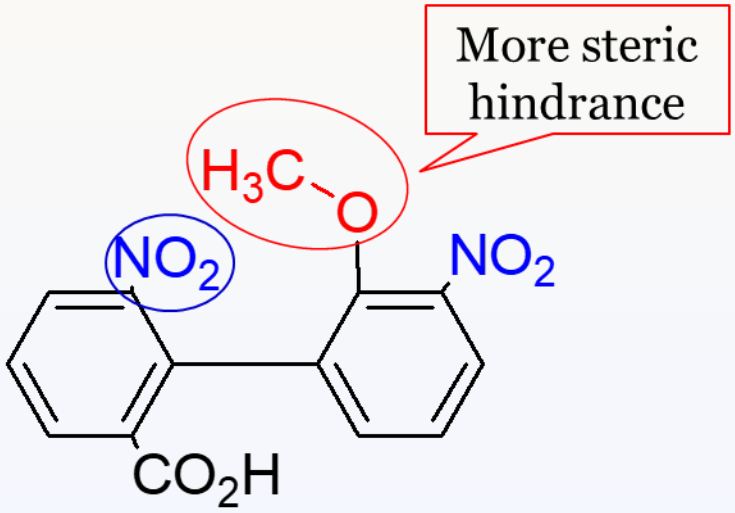
(II)

Buttressing Effect

- The methyl group in methoxyl group is probably in the (III) configuration.
- In (IV), the nitro group in the 3′-position would tend to force the methyl group away ⇒ Greater interference between the methoxyl group and two groups in other benzene ring.



(III) racemizes rapidly

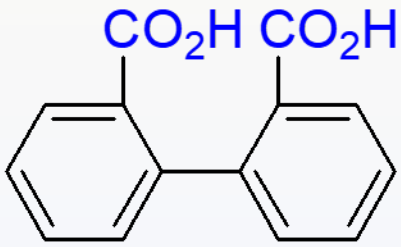


(IV) racemizes slowly

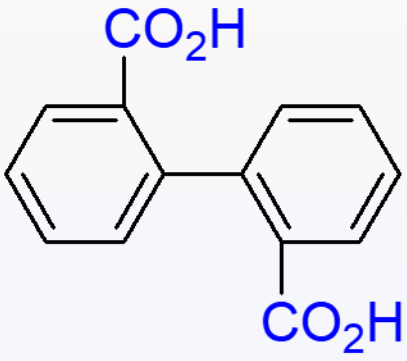
- The order of buttressing effect of groups: NO_2 > Br > Cl > Me

Bridged Biphenyls

- Diphenic acid [configurations (V) or (VI)] is not optically active. Its configuration is most probably (VI).
- However the compound (VII) can be resolved. The two benzene rings are not coplanar and are held fairly rigid by large methylene bridge.



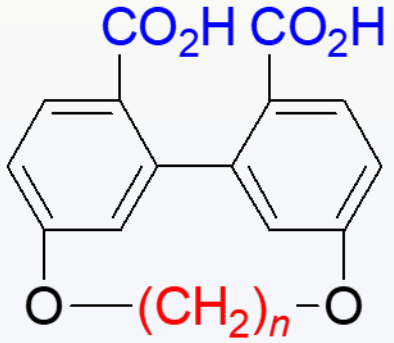
(V)



(VI)

Diphenic Acid

Optically Inactive



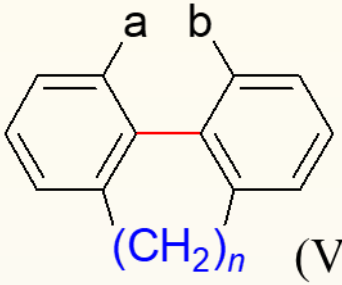
(VII)

$n = 8$ or 10

Optically Active

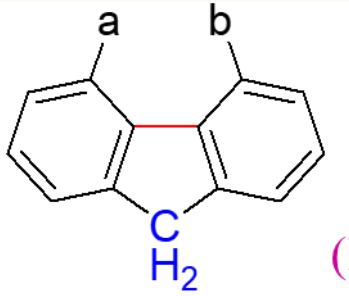
Bridged Biphenyls

- Many biphenyls have been studied from the point of view of the effect of a 2,2′-bridge on the optical activity of molecules of the type (VIII)



(VIII)

- When n = 1, the molecule is disubstituted fluorene (IX). Since this molecule is flat, it is not resolvable.

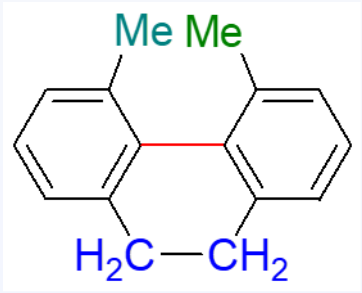


(IX)

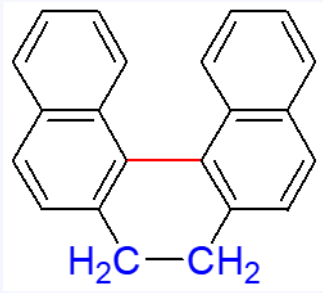
Optically Inactive

- When n = 2, the molecule is disubstituted 9,10-dihydrophenanthrene. Compounds like (X) and (XI) have been resolved.

- (XI) is more stable than (X). In (X), the two Me groups can slip past each other comparatively easily by bending out of the ring plane.



(X)

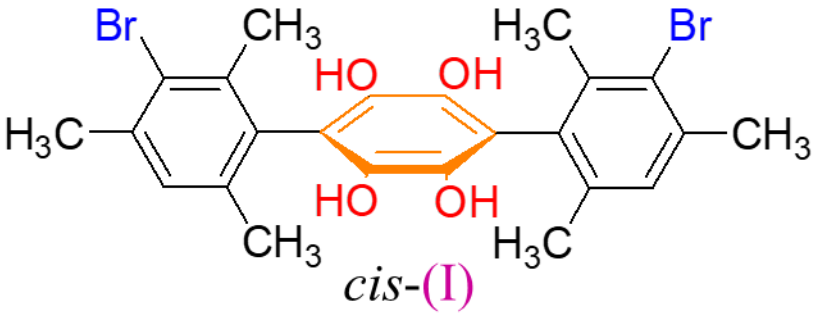


(XI)

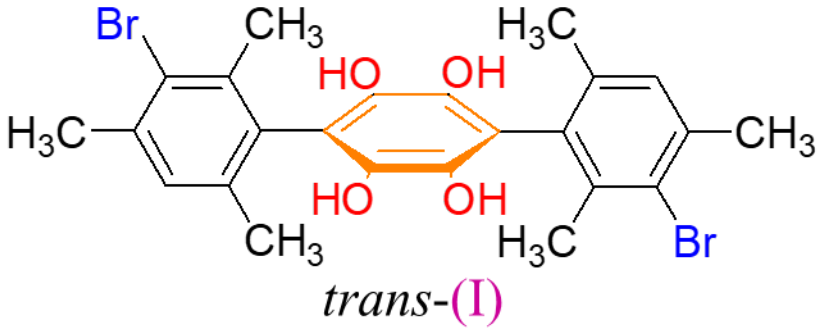
Optically Active

Terphenyl Compounds

- Terphenyl compounds can exhibit both geometrical and optical isomerism
- Interference of the methyl and hydroxyl groups in the *ortho*-positions of compound (I) prevents free rotation and tends to hold the two outside rings perpendicular to the central ring.

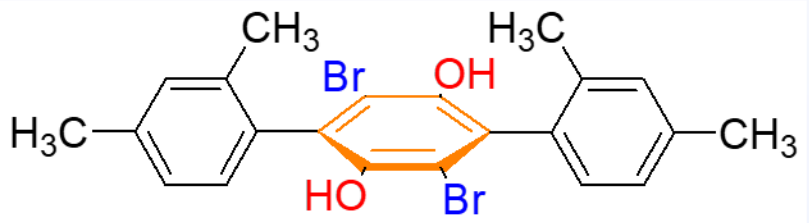


cis-(I)

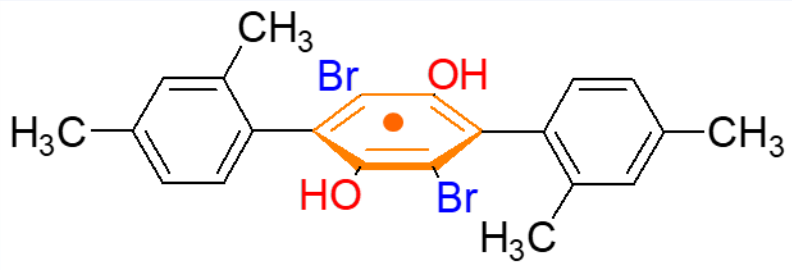


trans-(I)

- If the central ring does not possess a vertical plane of symmetry, then optical activity is possible.



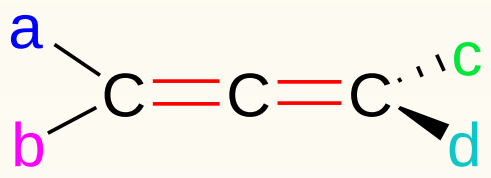
Optically active



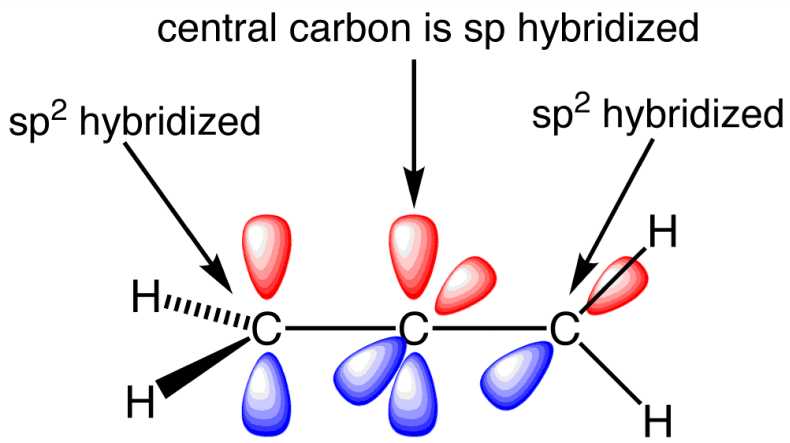
Not resolvable

Chirality in Molecules Devoid of Chiral Centers

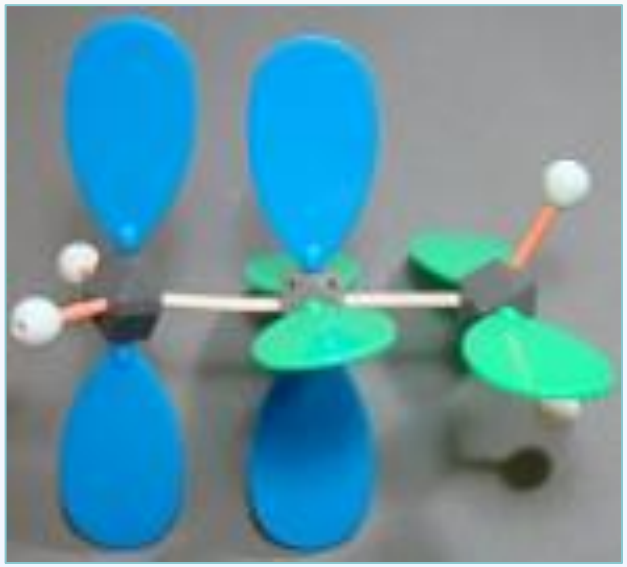
Allenes

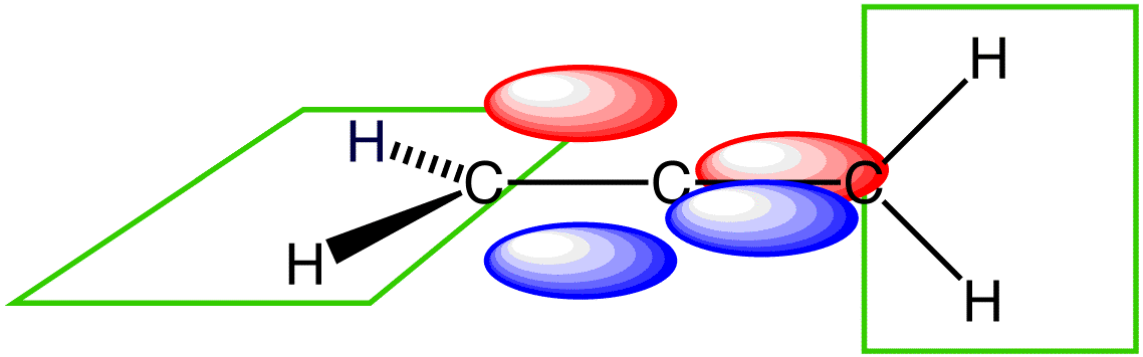


- Allenes are classified as polyenes with cumulated dienes, *i.e.*, in allene one carbon atom has double bonds with each of its two adjacent carbon centers.



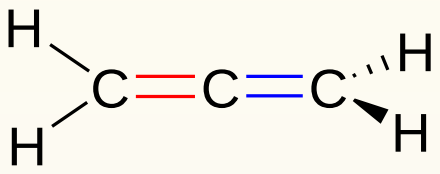
the π bonds formed as a result of the overlap of the p orbitals must be at right angles to each other

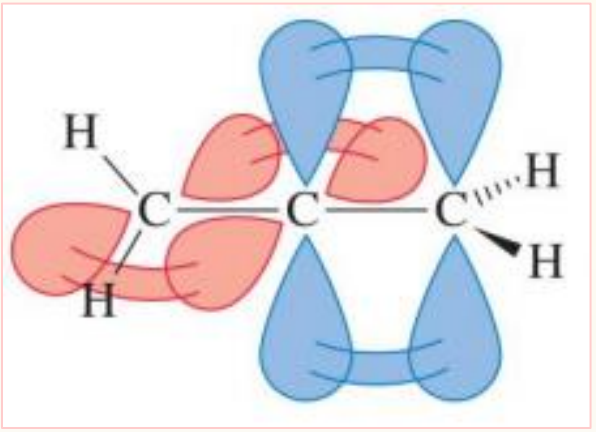




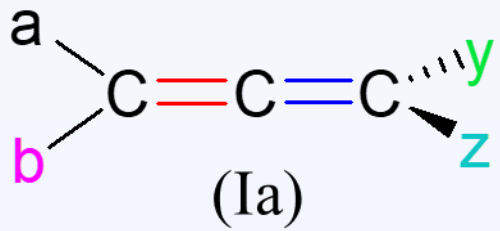
not only are the two π bonds perpendicular, but the two methylene groups are too

Allenes

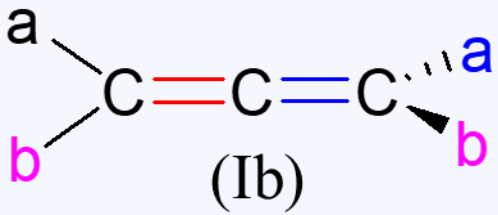




- 1,2-Propadiene is the simplest allene.
- Allene itself is achiral since it has plane of symmetry.
- In order to generate chirality, the plane of symmetry must be eliminated.
- When different substituents are added at each end of the C=C=C unit, the substituted allene (Ia) or (Ib) becomes chiral.

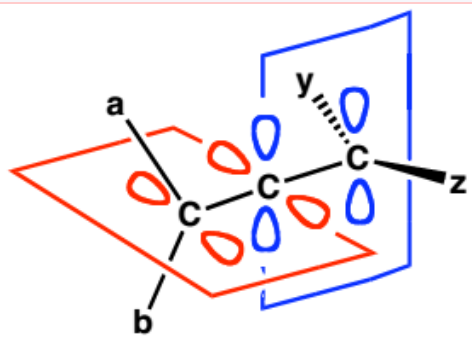


(Ia)



(Ib)

✓ Condition for chirality is a ≠ b and y ≠ z.



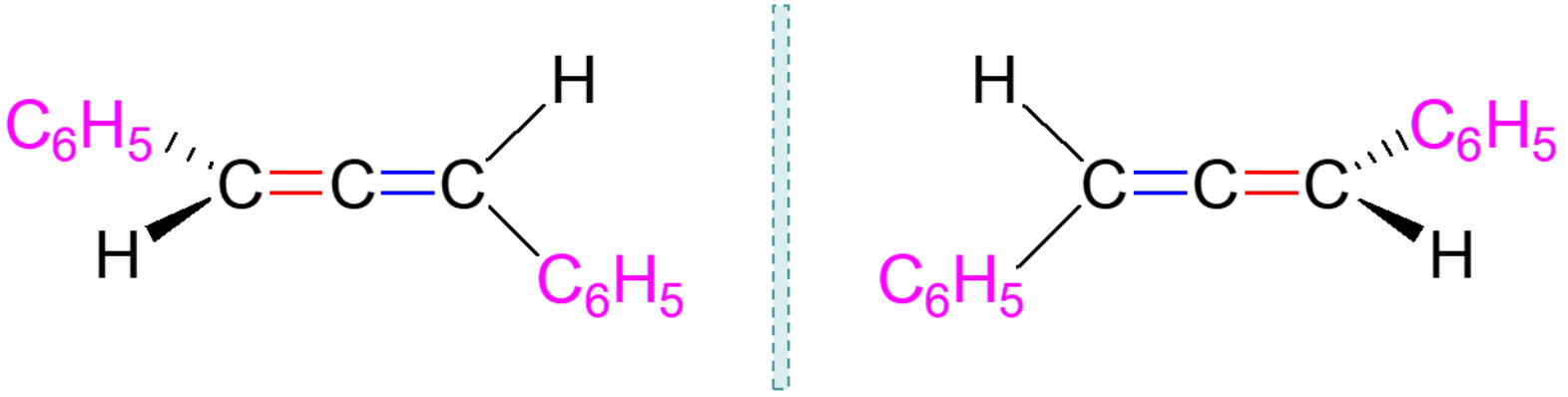
Orthogonal Bonding

Optically Active Allenes

- Since the geometry of the cumulated π-bonds causes the groups at the terminal carbons to lie in perpendicular planes, allenes with different substitutents attached to the end carbons form non-superimposable mirror images and they show optical activities even though they do not have a chiral carbon.





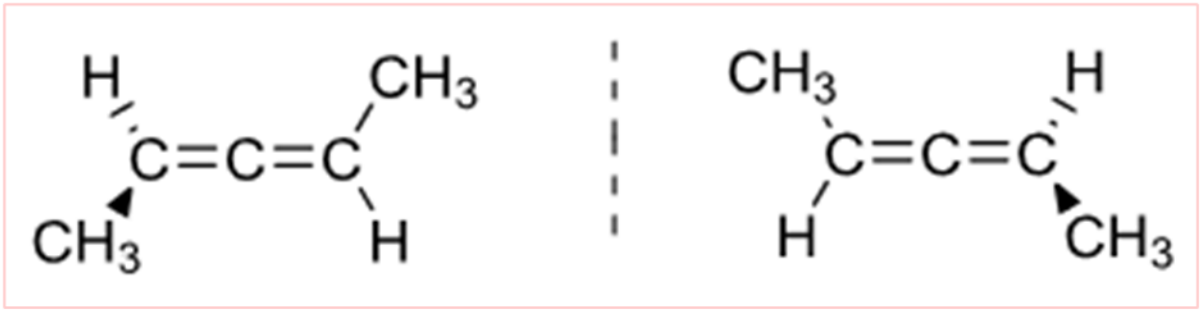


1,3-Diphenyl propadiene

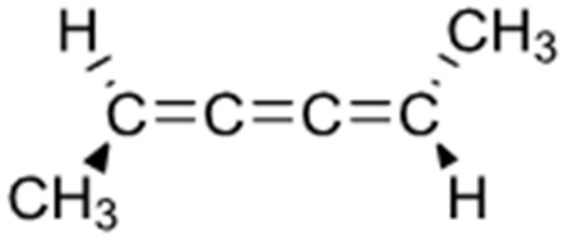
Optically Active Allenes

- Allenes with even number of cumulated double bonds are optically active if the groups present on the terminal carbons are different ⇒ this is due to lack of plane of symmetry and centre of symmetry in the molecule.

- However, allenes with odd numbers of cumulated double bonds are not chiral ⇒ they display *E*-*Z* (geometrical) isomerism.



Optically active



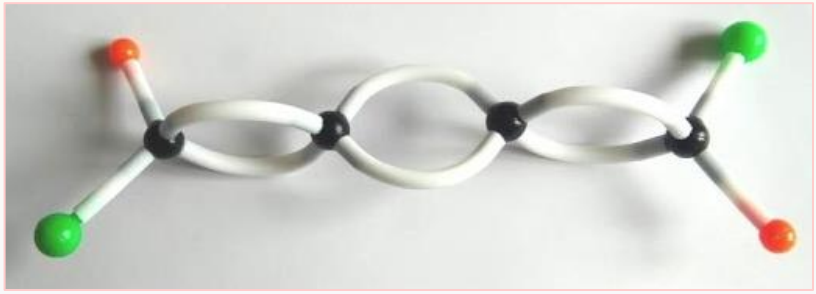
Achiral

Cumulene
(odd number of double bonds)

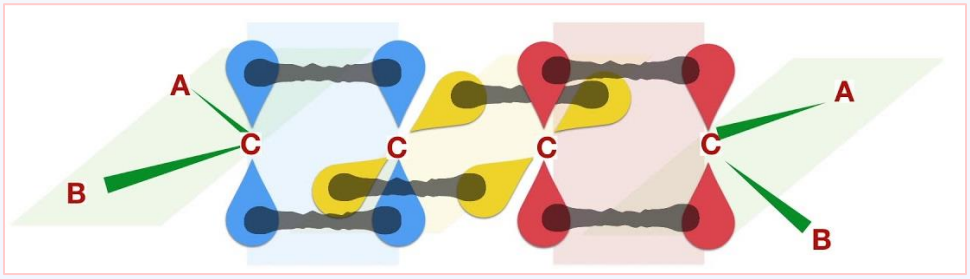
- The compounds with odd numbers of cumulated double bonds show geometrical isomerism.
- When the allene chain is extended by one more double bond, then the substituents at the two ends of the cumulated chain now lie in the same plane and show geometrical isomerism.

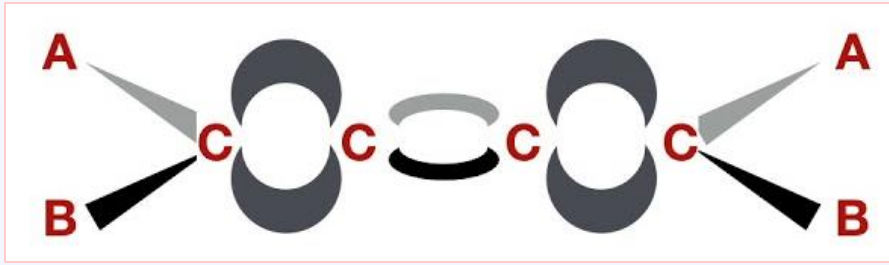


Allene
(with even no. of double bonds)



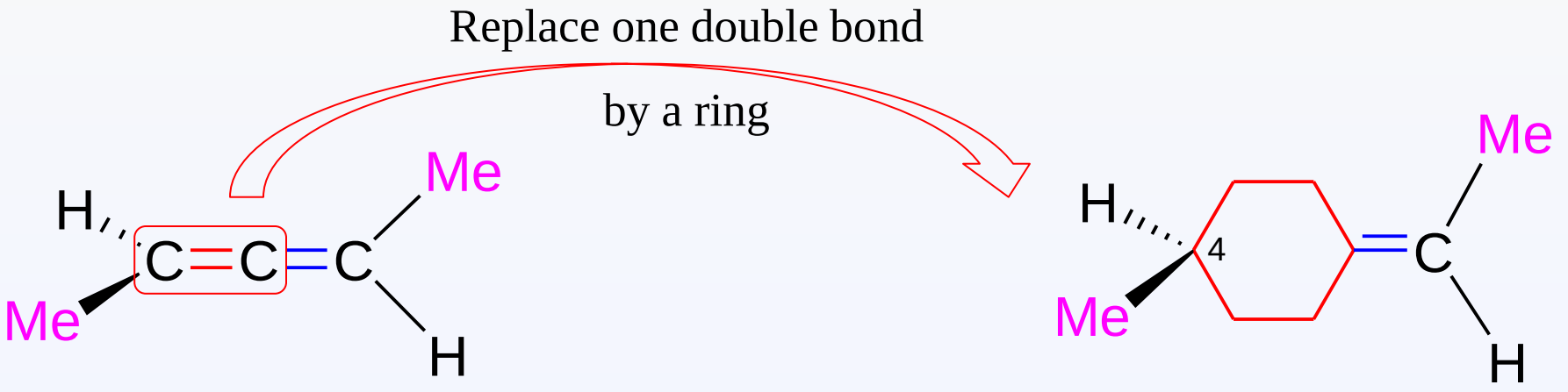
Cumulene
(with odd no. of double bonds)





Alkylidene Cycloalkanes (Hemispiranes)

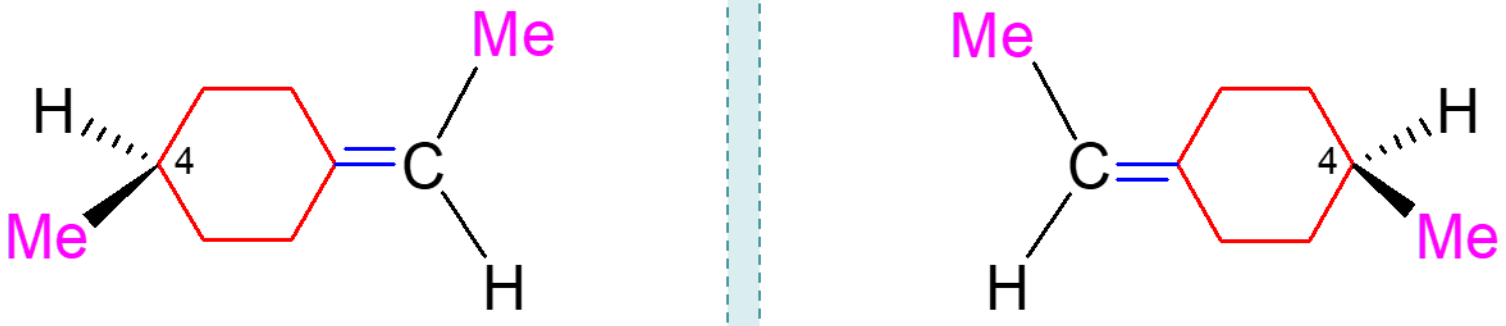
- The replacement of one double bond in an allene by a ring gives alkylidene cycloalkane.
- The basic geometry of the system of allenes is not altered by the replacement of the double bond by a ring.
- So, the alkylidene cycloalkane can exist in optically active form when the substituents at the terminal carbons are different.



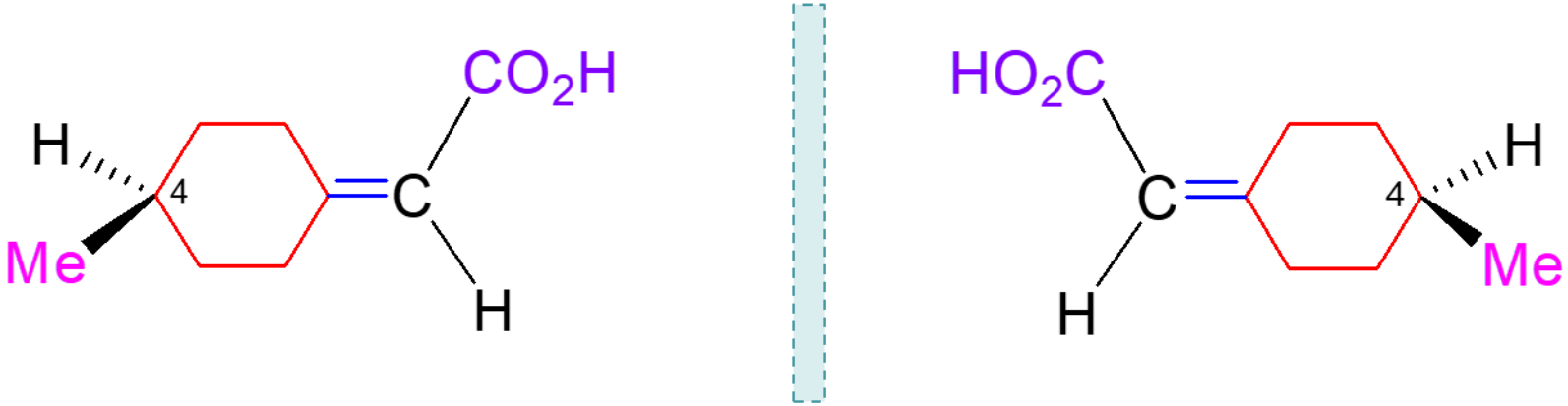
Chiral Allene

Chiral alkylidene cycloalkane

Alkylidene Cycloalkanes



Non-superimposable Mirror Images

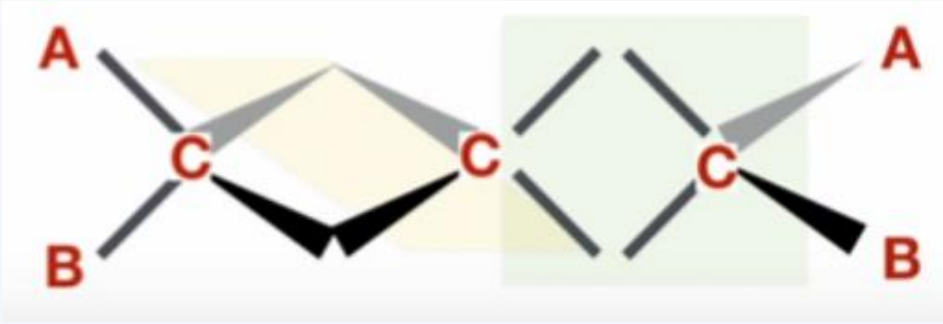


Non-superimposable Mirror Images

Spiranes

- If both double bonds in an allene are replaced by ring systems, the resulting molecules are spiranes.
- So, spiranes are bicyclic compounds in which one carbon atom (sp^3 hybridized) is common to both the rings
- In spiranes, the two rings are perpendicular to each other, and hence unsymmetrical substitution at the two ends will produce molecules with no elements of symmetry ⇒ optically active spirane molecules.

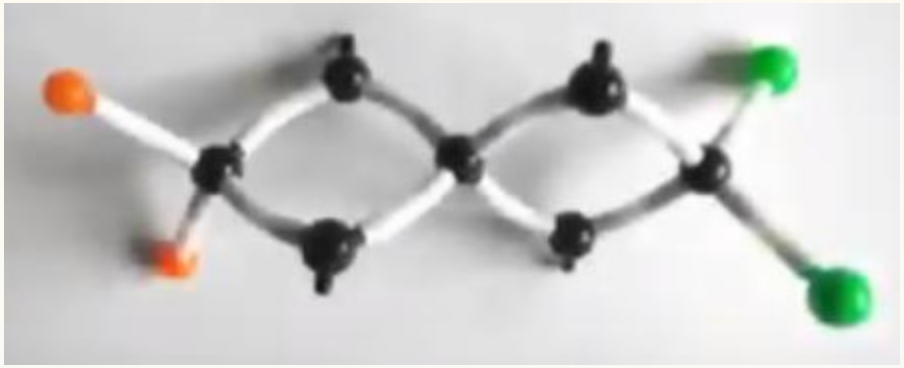


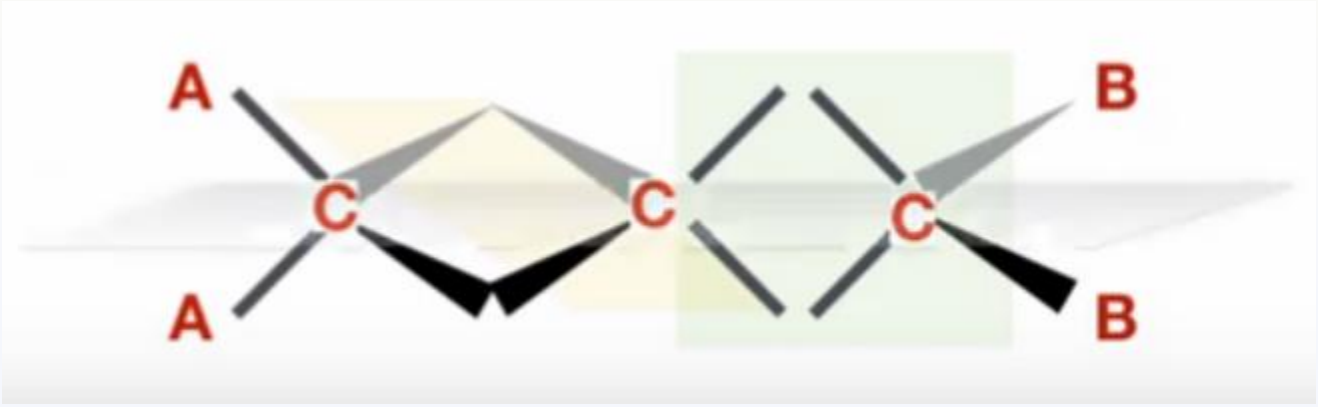


✓ Condition for chirality is A ≠ B

Optically Inactive Spiranes

- If the substituents at each end of the spiran molecule are the same, then there exist plane of symmetry and the compound becomes optically inactive.



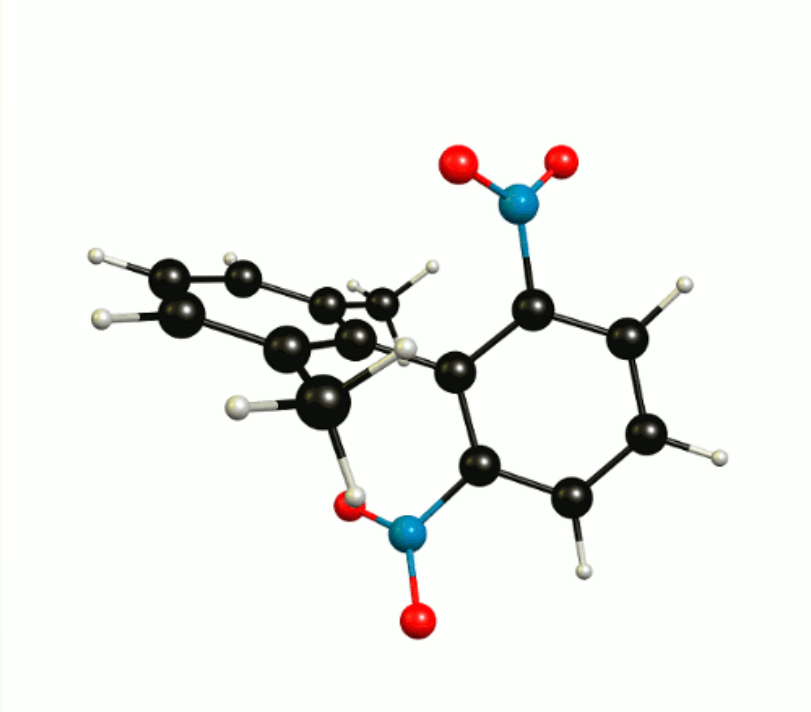


Spirane with same substituents on terminal carbon atoms

Plane of symmetry exists

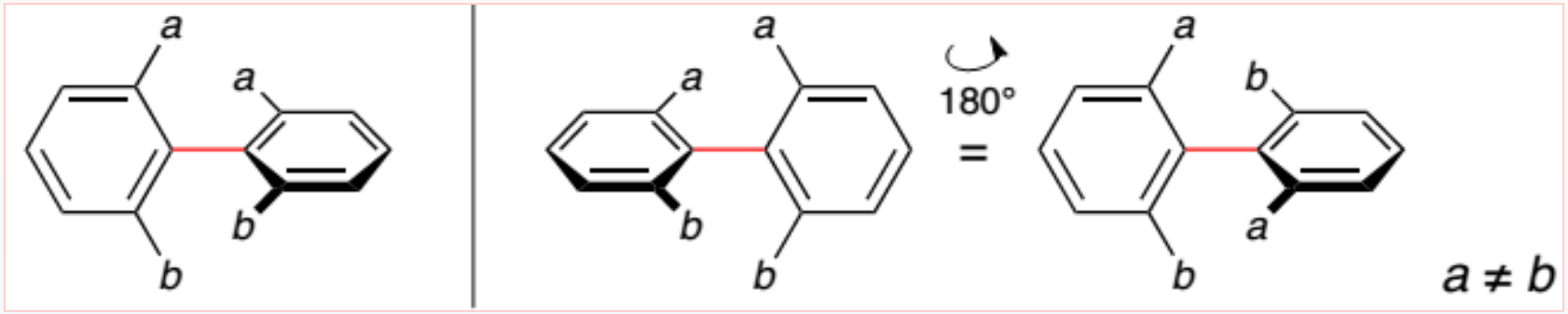
Optically Inactive





Optical Activity Of Biphenyl Compounds

Chiral Molecules Without Chiral Centers



Optically active biphenyl compound

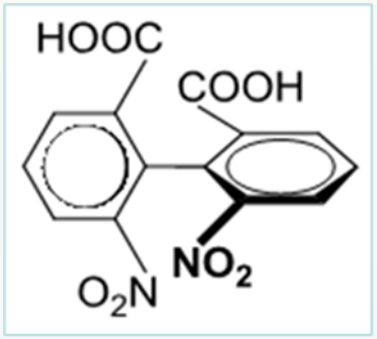
Non-superimposable with its mirror image

Atropisomerism



Richard Kuhn
(1900-1967)
Nobel **Laureate**
(1938)

- The term "Atropisomerism" was coined by Richard Kuhn in 1933.
- However "Optical activity due to axial chirality" was first reported by Christie and Kenner in 1922. Axial chirality was detected in 6,6'-dinitro-2,2'-diphenic acid.

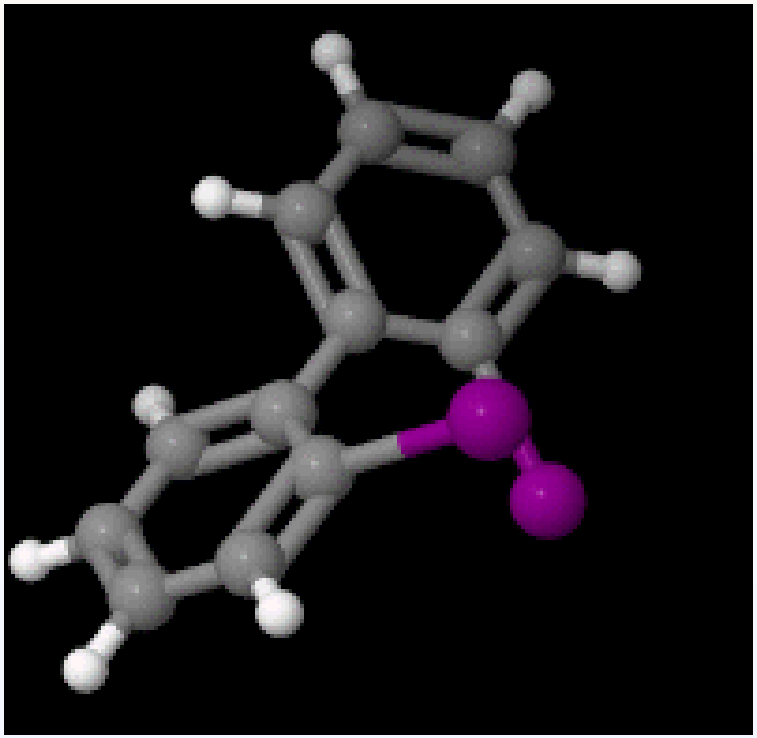


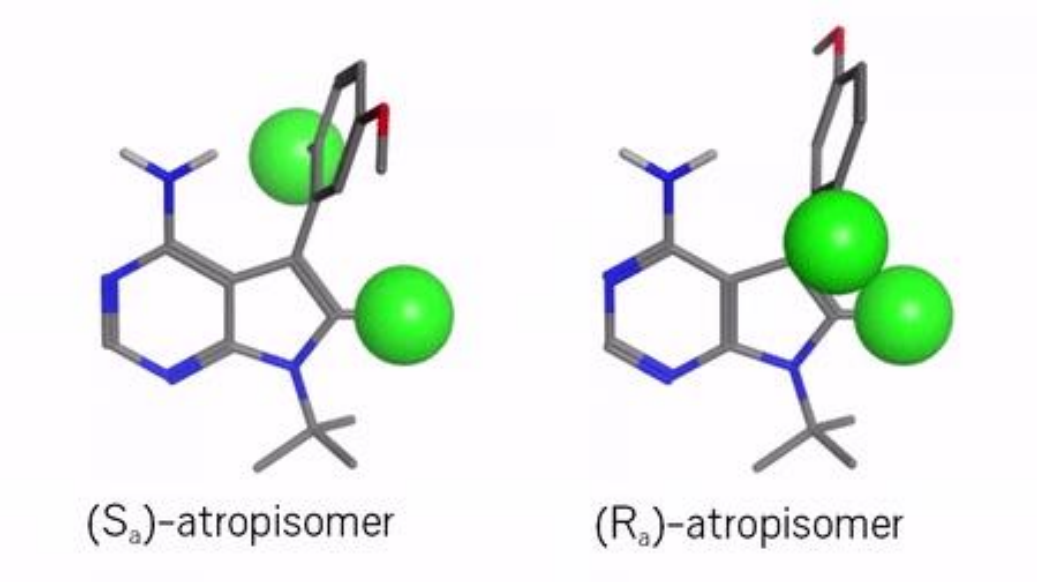
Single isomer was isolated from the racemic mixture *via* diastereoselective crystallization with a chiral resolving agent

- Optical isomers produced due to restricted rotation about a single bond is called atropisomers.
- Restricted rotation occurs when *o*-positions contain *two different bulky groups* and hence molecule is optically active.

Atropisomerism

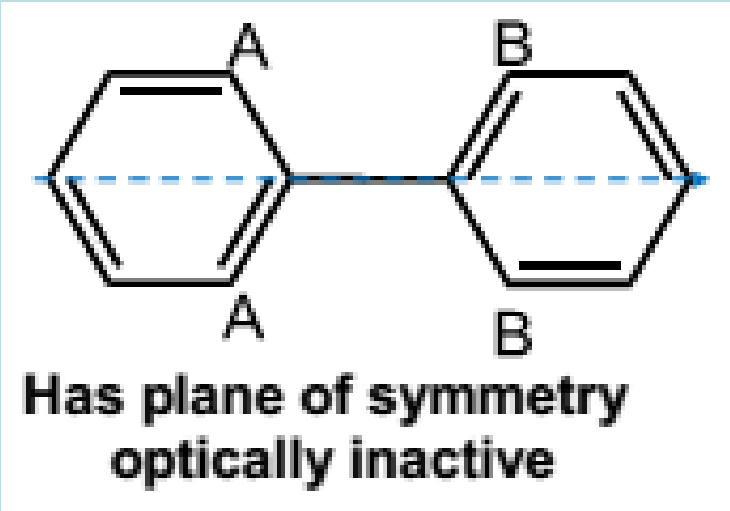
- **Atropisomerism** occurs when there is hindered rotation about a single bond as a result of steric or electronic constraints.



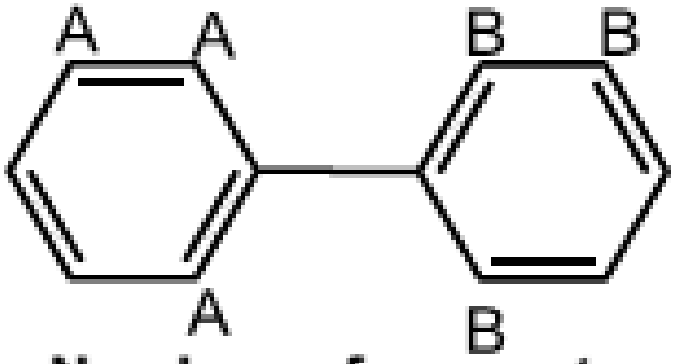


Optical Activity Of Biphenyl Compounds

- Conditions to exhibit optical activity for biphenyl compounds:
  - Neither ring must have a plane of symmetry.
  - Ortho-positions must be occupied by large groups or atoms.

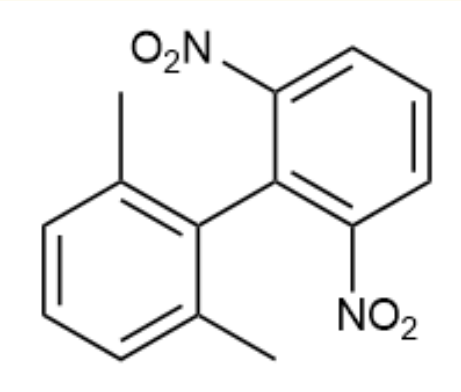


Has plane of symmetry
optically inactive

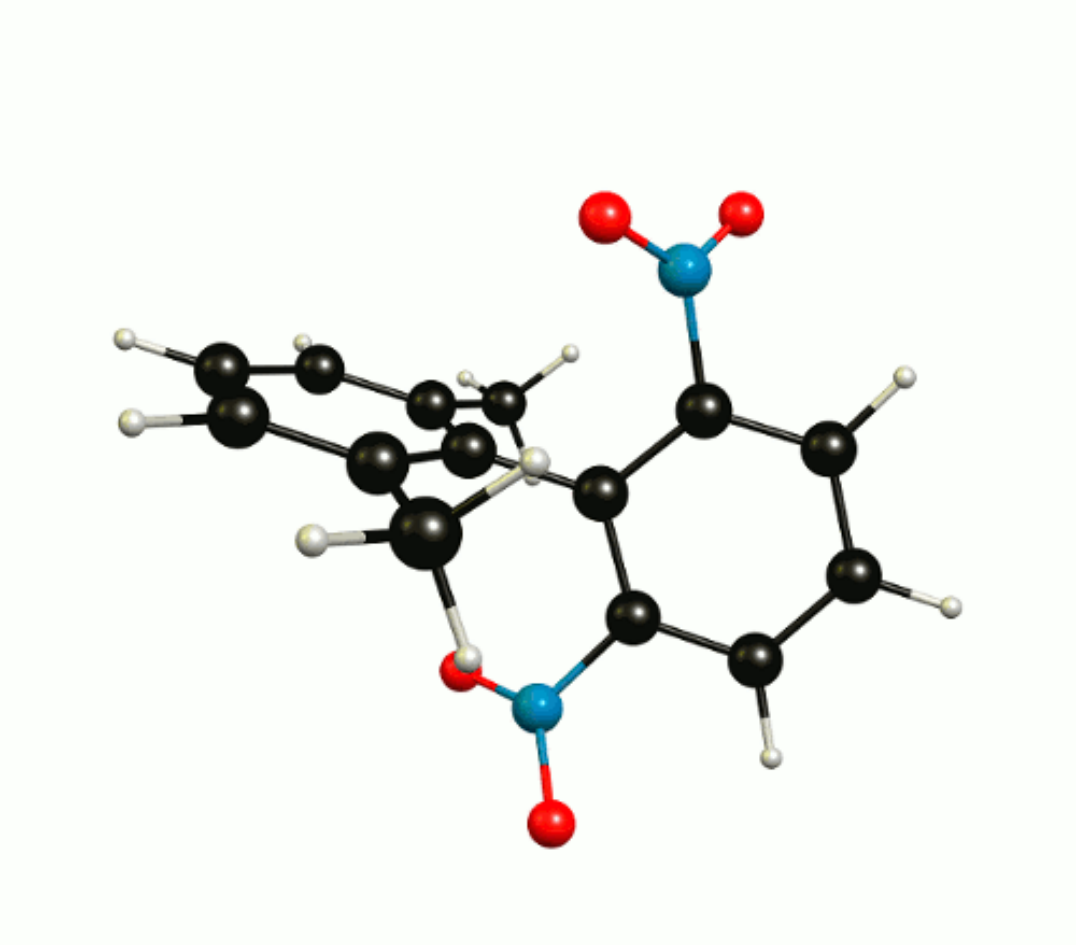


No plane of symmetry
optically active,

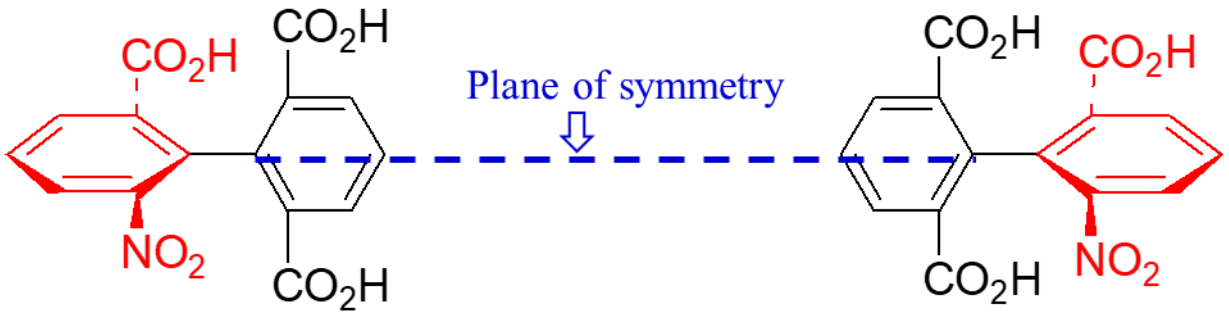
Optical Activity Of Biphenyl Compounds



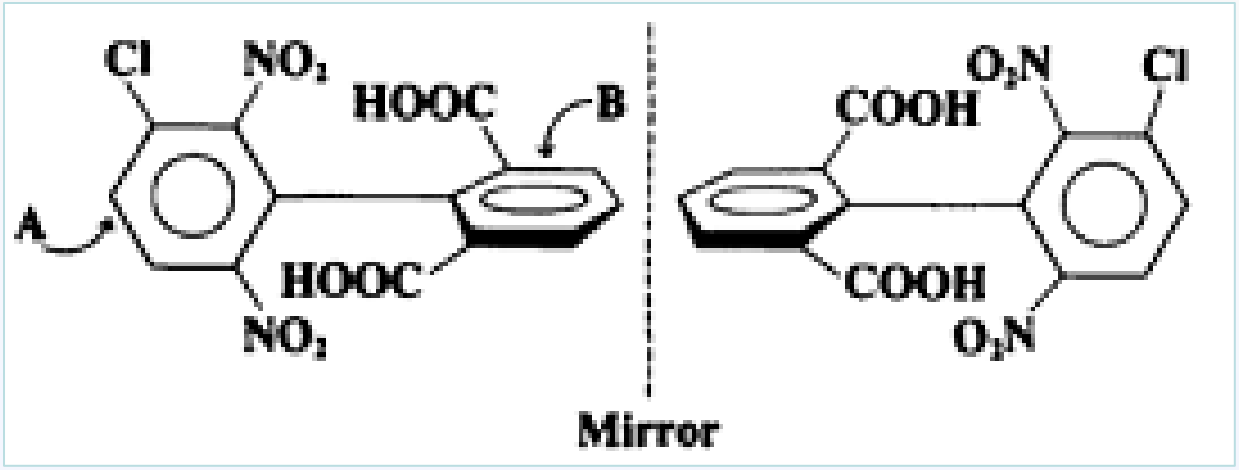
2,6-dimethyl-2',6'-dinitro-1,1'-biphenyl



Optical Activity Of Biphenyl Compounds



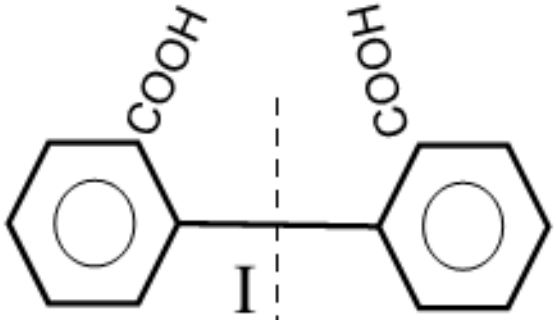
- Optically inactive due to presence of plane of symmetry



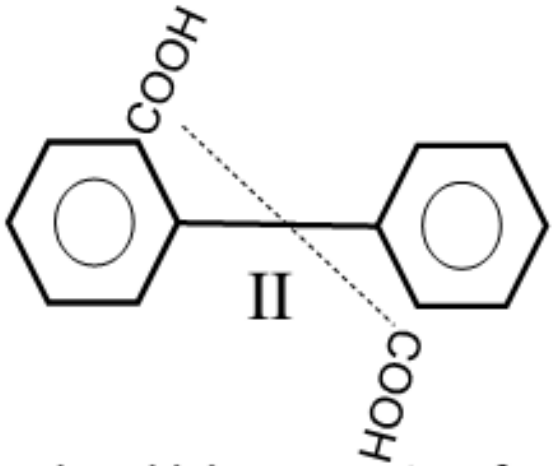
- Ring B is symmetrically substituted. A plane of symmetry exists.

The compound is achiral

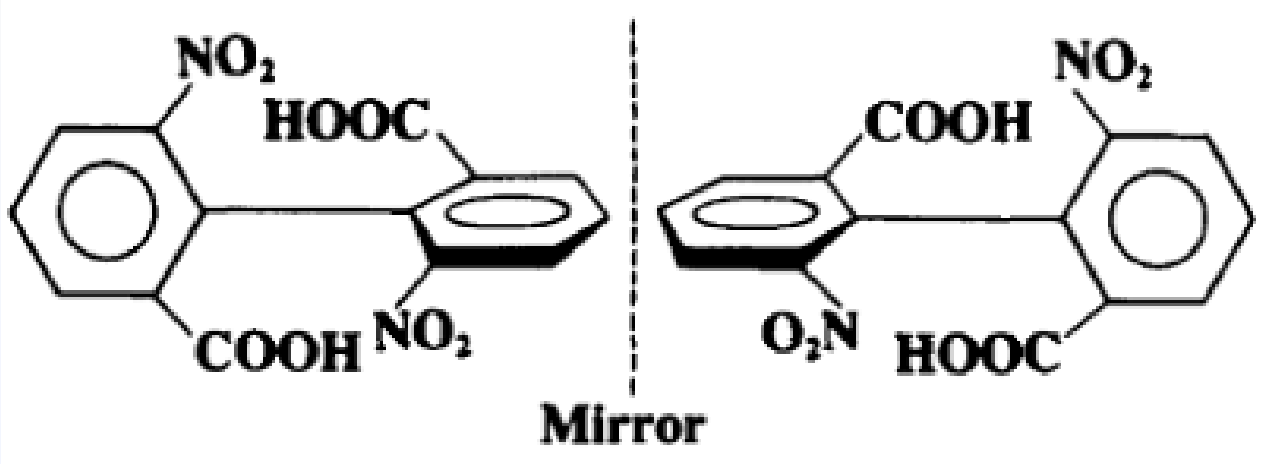
Optical Activity Of Biphenyl Compounds



Diphenic acid, has a plane of symmetry
Optically inactive



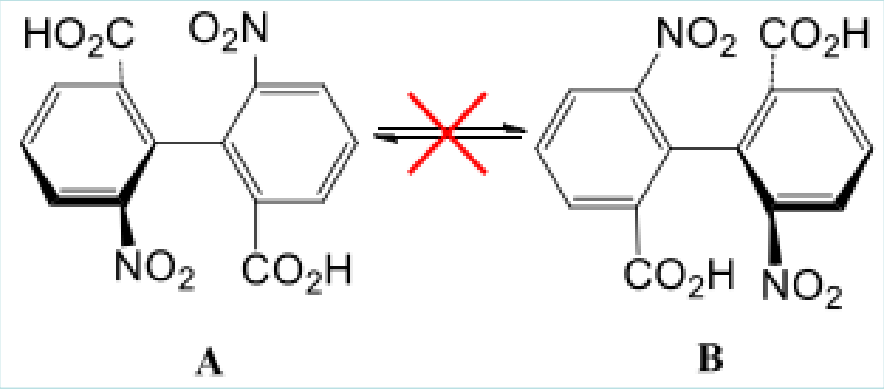
Diphenic acid, has a centre of symmetry
Optically inactive

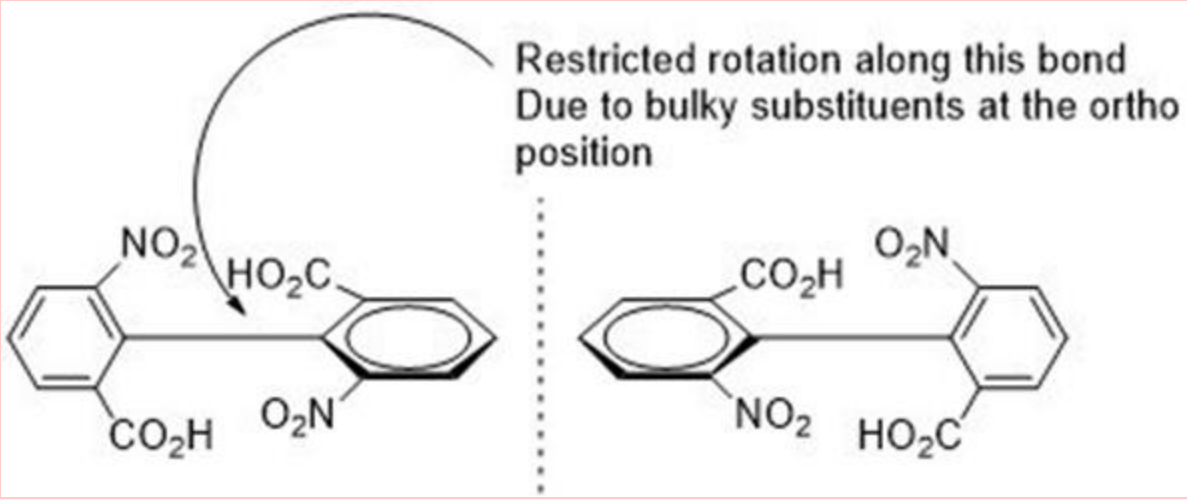


- No plane of symmetry, so optically active.

Optical Activity Of Biphenyl Compounds

- Conformers that cannot interconvert (due to steric hindrance) can be enantiomers.

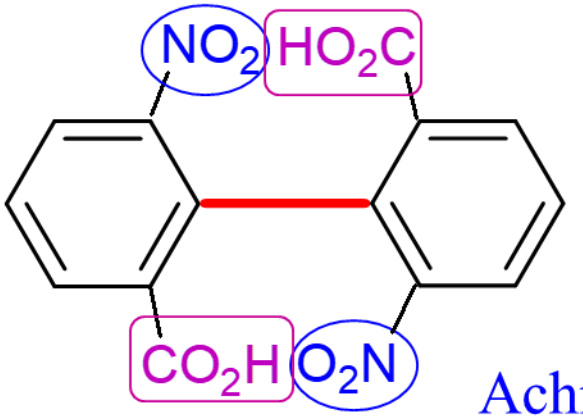




Non-coplanar, optically active

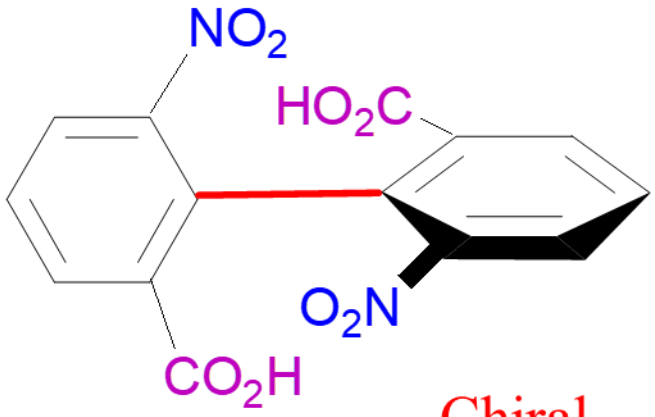
Atropisomers

Impossible, too crowded

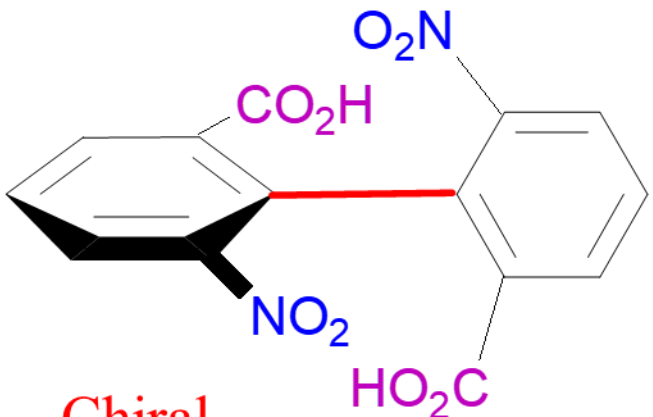


Achiral

Due to the presence of large *ortho*-substituents, two benzene rings can not be coplanar



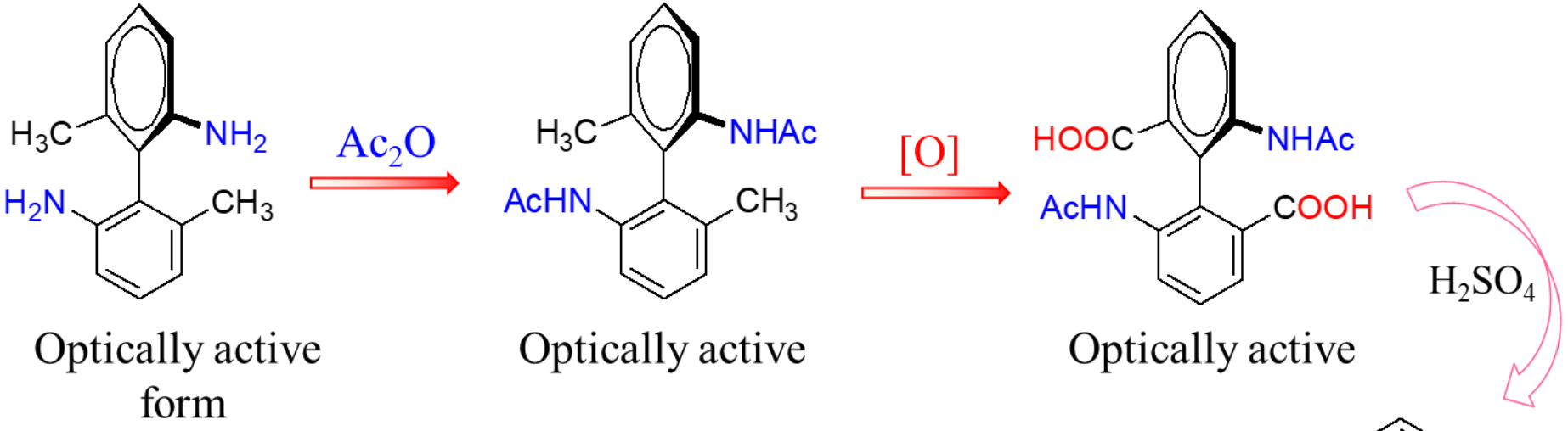
Chiral



Chiral

- In optically active biphenyl compounds, two rings tend to be perpendicular.

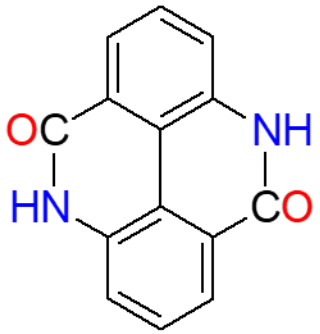
Evidence for Obstacle Theory



Optically active form

Optically active

Optically active



Center of symmetry present

Optically inactive

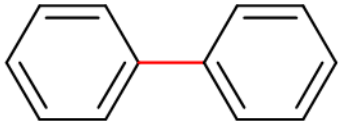
- Evidence for the obstacle theory, *i.e.*, interference of groups, amounts to proving that the two rings in optically active compounds are not coplanar.
- In all the optically active compounds, the rings can not be coplanar.
- The dilactam is planar and has a center of symmetry, so it is optically inactive

Evidence for Obstacle Theory

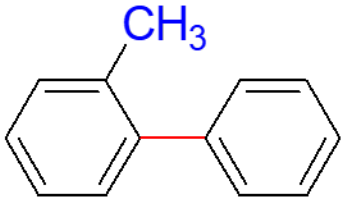
- Ultraviolet spectra measurements of biphenyl compounds have shown that two rings in *o*-substituted biphenyls are not coplanar.



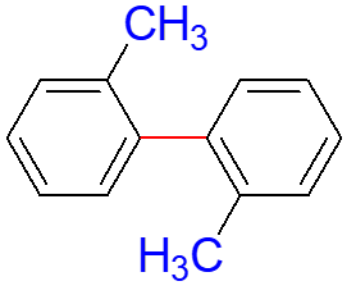
$\lambda_{max} = 198$ nm
(ε 8000)



$\lambda_{max} = 248$ nm
(ε 19000)



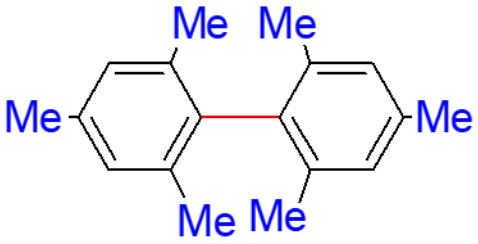
$\lambda_{max} = 236$ nm
(ε 10000)



$\lambda_{max} = 224$ nm
(ε ~ 700)

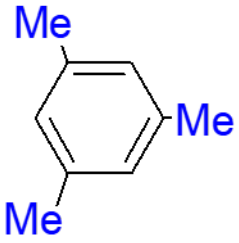
- When *o*-positions are occupied, the coplanarity is prevented, the maximum wavelength will shit to shorter wavelength.

- UV spectra of bimesityl and mesitylene are almost identical. In (I) coplanarity is prevented due to steric inhibition of resonance.



(I)

$\lambda_{max} = 267$ nm (ε 545)

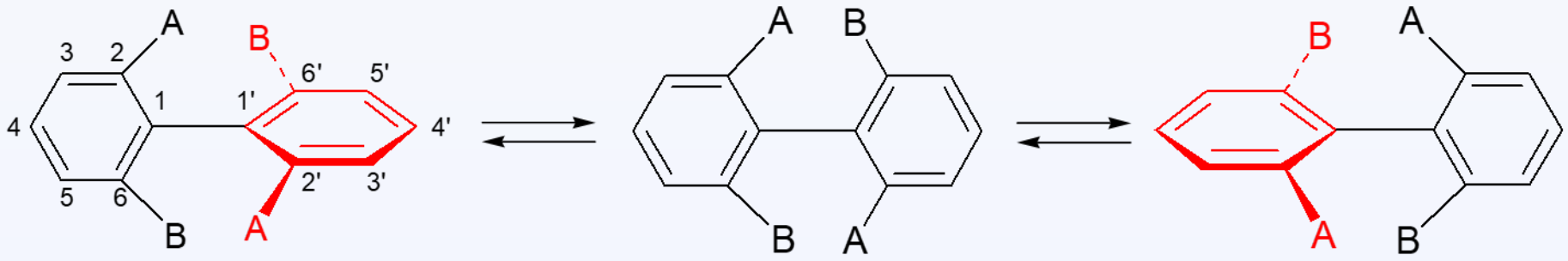


(II)

$\lambda_{max} = 266$ nm (ε 260)

Racemization of Biphenyl Compounds

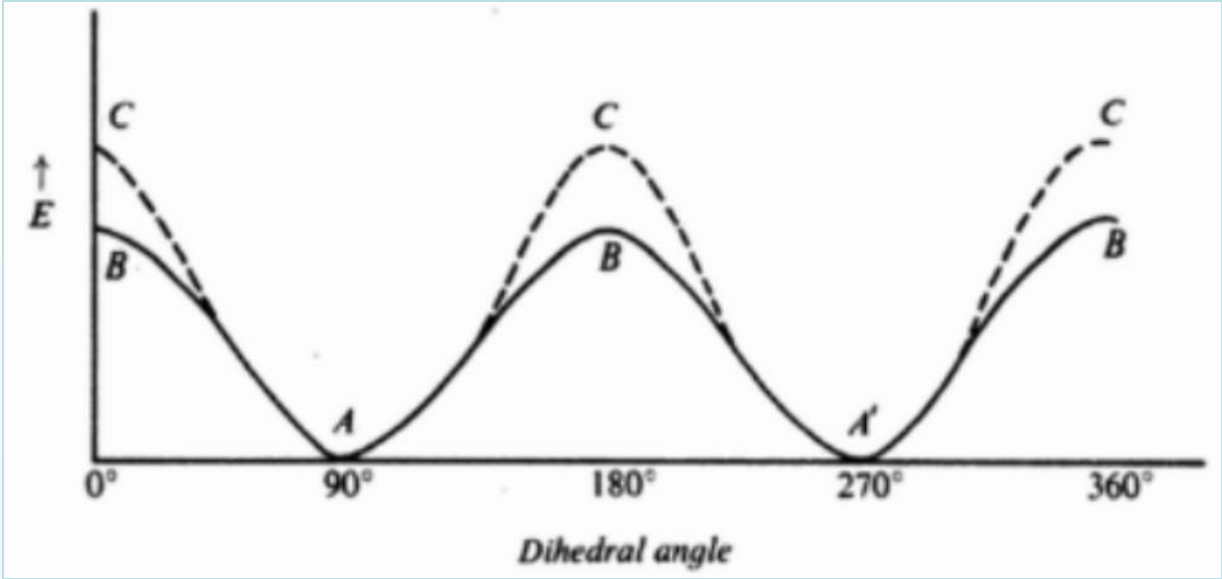
- Since the optical activity of biphenyl compounds arises from restricted rotation, it might be expected that racemization of these compounds would not be possible.
- In practice, many optically active biphenyl compounds can be racemized under suitable conditions, *e.g.*, boiling in solution.
- Heating increases the amplitude of the vibrations of the substituent groups in the 2,2′,6,6′-positions, and also the amplitude of vibration of the two benzene rings with respect to each other, thereby permitting the substituent groups to slip by one another.
- Thus the nuclei pass through a common plane. So the final product will contain equimolecular amount of the (+)- and (–)-forms.

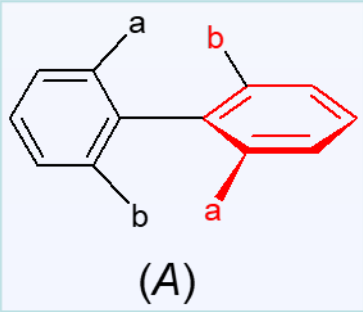


Planar form

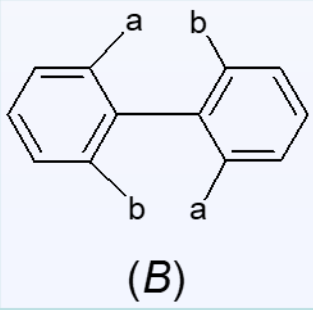
Racemization of Biphenyl Compounds

- The enantiomers (*A*) and (*A′*) have the same energy content. As they approach the planar configuration (B) the energy content increases due to steric repulsion.
- The larger the groups a and b, the higher will be the energy barrier, *i.e.*, the slower will be the rate of racemization.

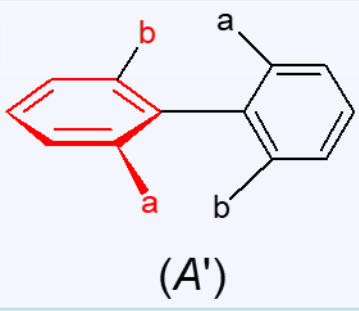




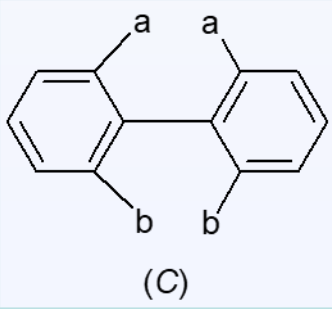
(*A*)



(*B*)



(*A'*)



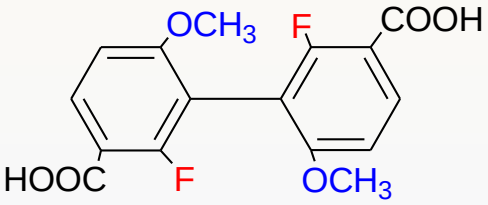
(*C*)

Racemization of Biphenyl Compounds

- 2,2′,6,6′-Tetrasubstituted biphenyl compounds may be classified under three headings according to the nature of the substituent groups:

i. Non-resolvable: These contain any of the groups (H, -OCH_3 or F).

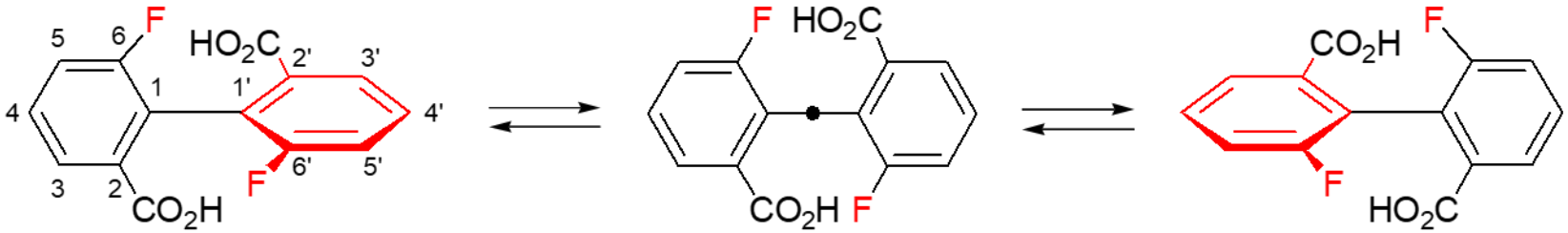
The effective volumes of these groups are too small to prevent rotation about the single bond.



Non-resolvable

ii. Resolvable but Easily Racemized: These must contain at least two –NH_2 or –COOH groups, or one –NH_2 and one –COOH group.

The remaining groups may be any of -OCH_3 or F , but not H.



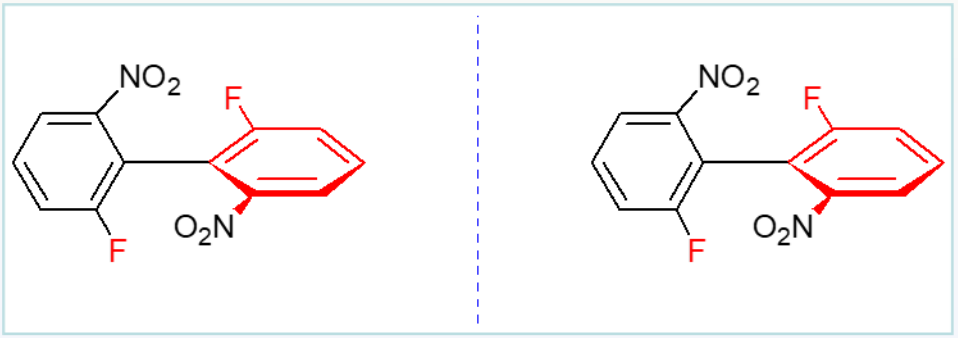
Planar form has a center of symmetry

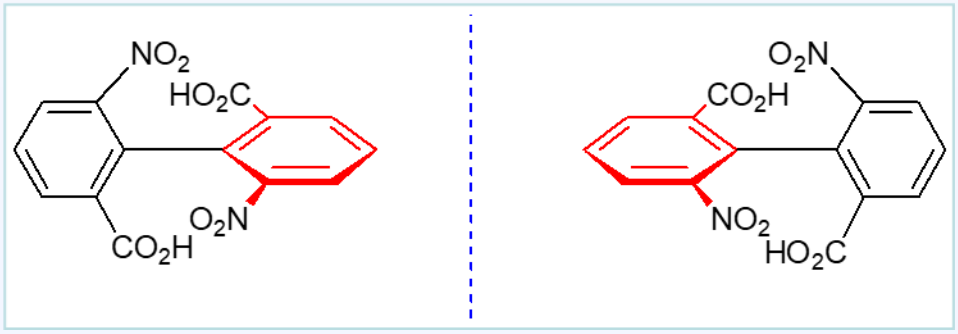
Racemization of Biphenyl Compounds

- 2,2′,6,6′-Tetrasubstituted biphenyl compounds may be classified under three headings according to the nature of the substituent groups:

iii. Not Racemizable at All: These biphenyl compounds contain least two $-NO_2$ groups.

The remaining groups may be any of $-OCH_3$, F, $-NH_2$ or –COOH groups but not H.

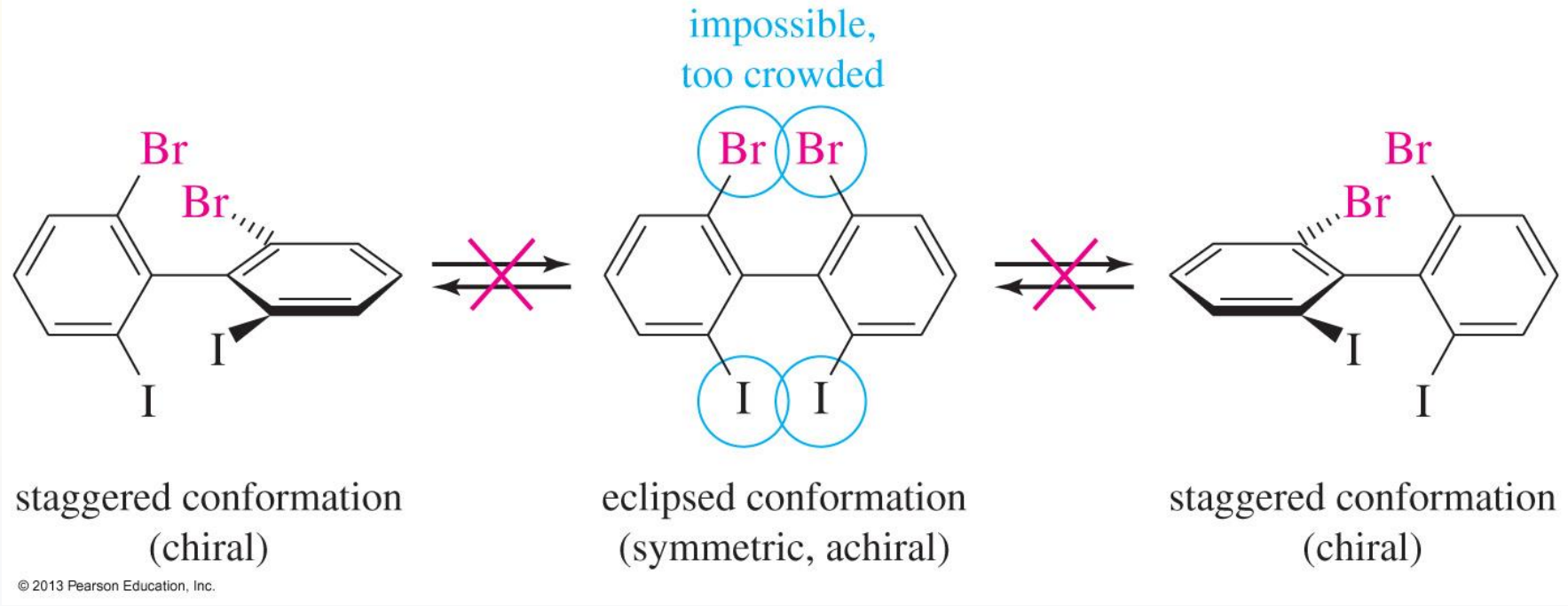




Resolvable and cannot be racemized.

Optically Active

Optical Activity of Biphenyl Compounds



The order of steric hindrance produced by various groups:

$$Br >> Me > Cl > NO_2 > CO_2H >> OMe > F$$

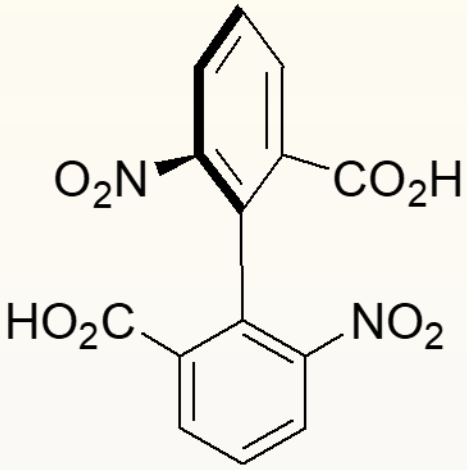
- This order roughly parallels van der Walls radii.

Chirality in Molecules Devoid of Chiral Centers

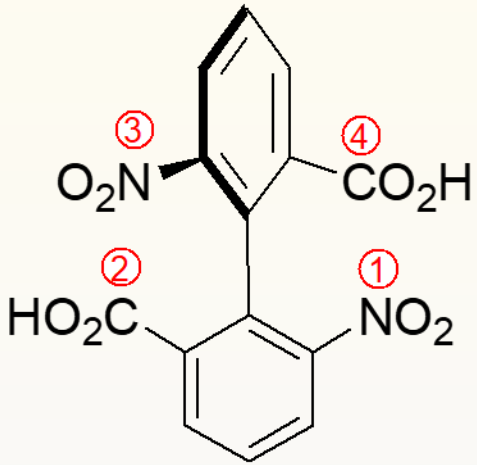
Assignment Of Configurational Nomenclature To Axially Chiral Molecules

- The molecules can be viewed from either of the stereoaxis to give the same descriptor.
- The near groups are given priority over far groups.
- For better visualization it is always better to put the groups nearest the view direction on thick line (horizontal or vertical).

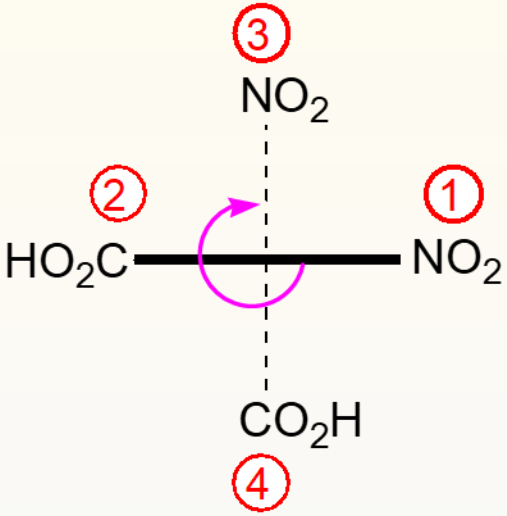
R/S Configuration (Biphenyls)



≡

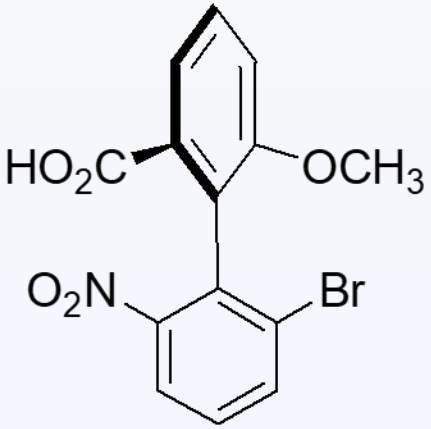


≡

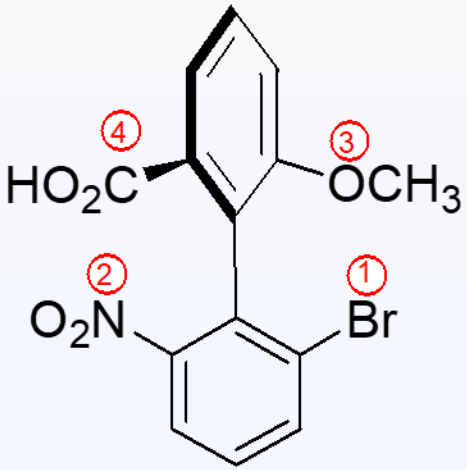


R-configuration

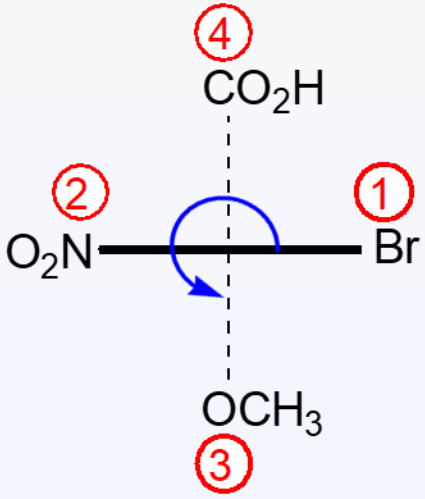
R-configuration



≡



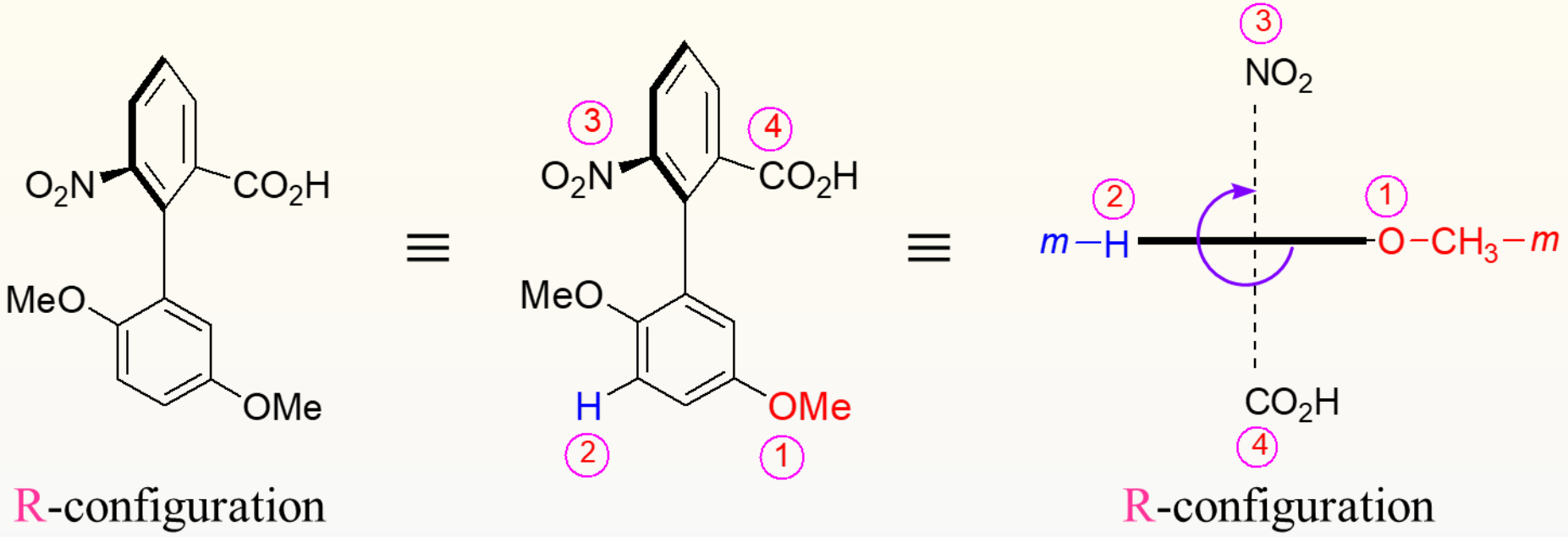
≡



S-configuration

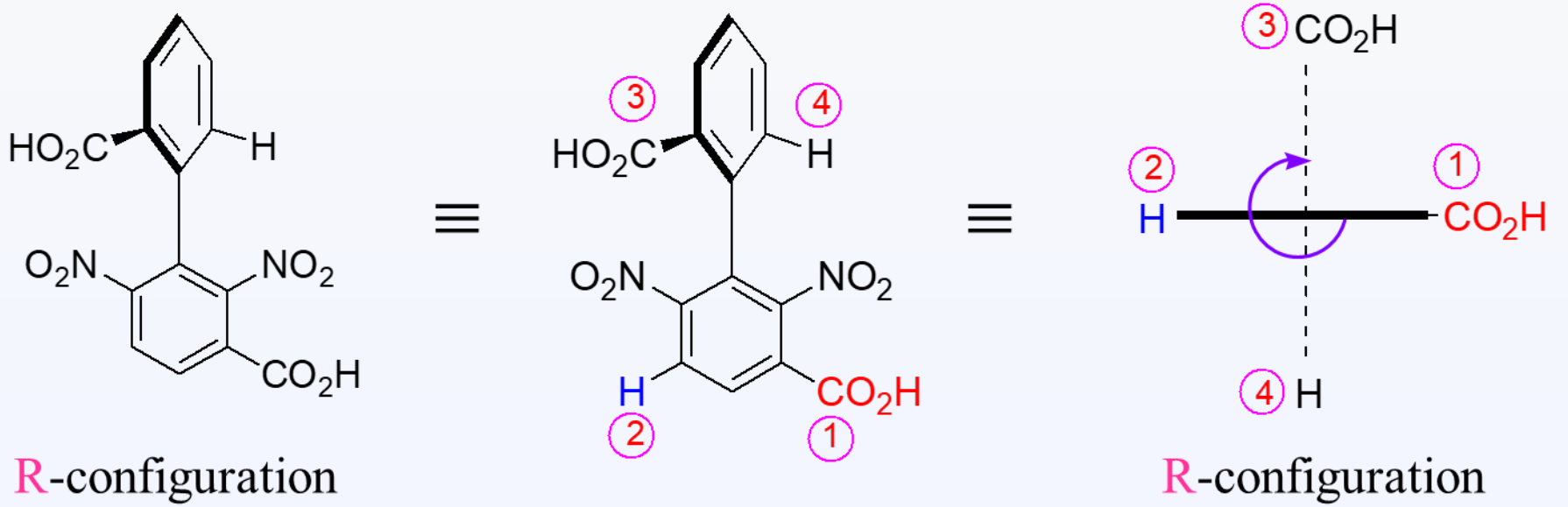
S-configuration

R/S Configuration (Biphenyls)



R-configuration

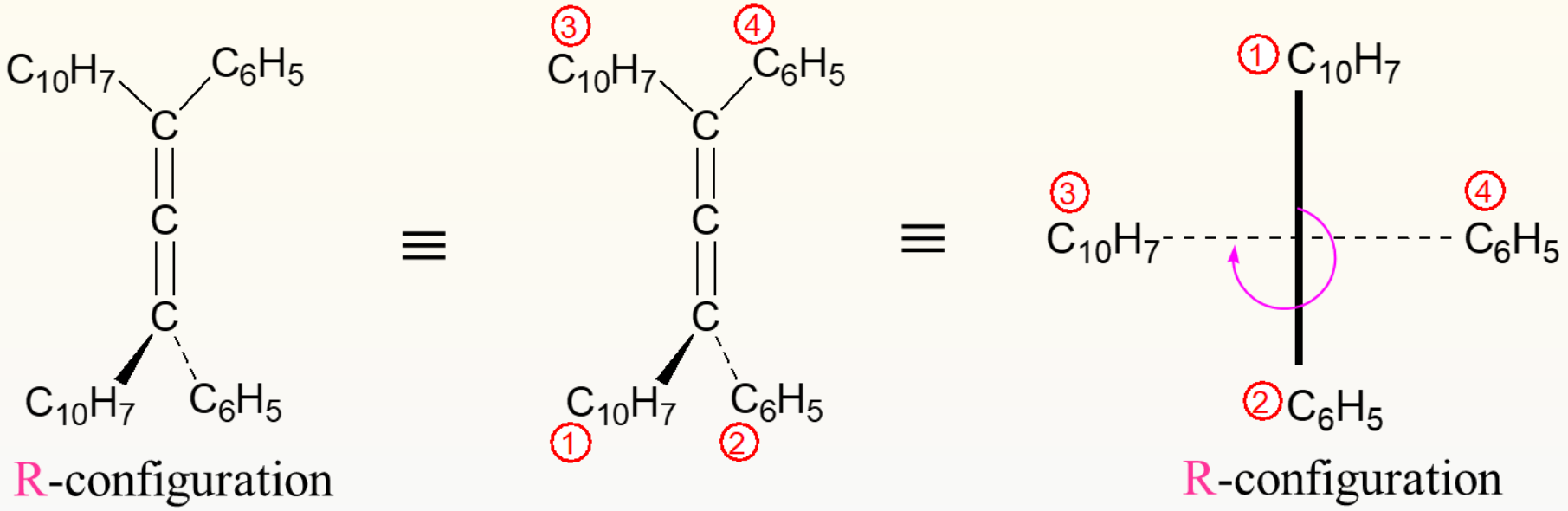
R-configuration

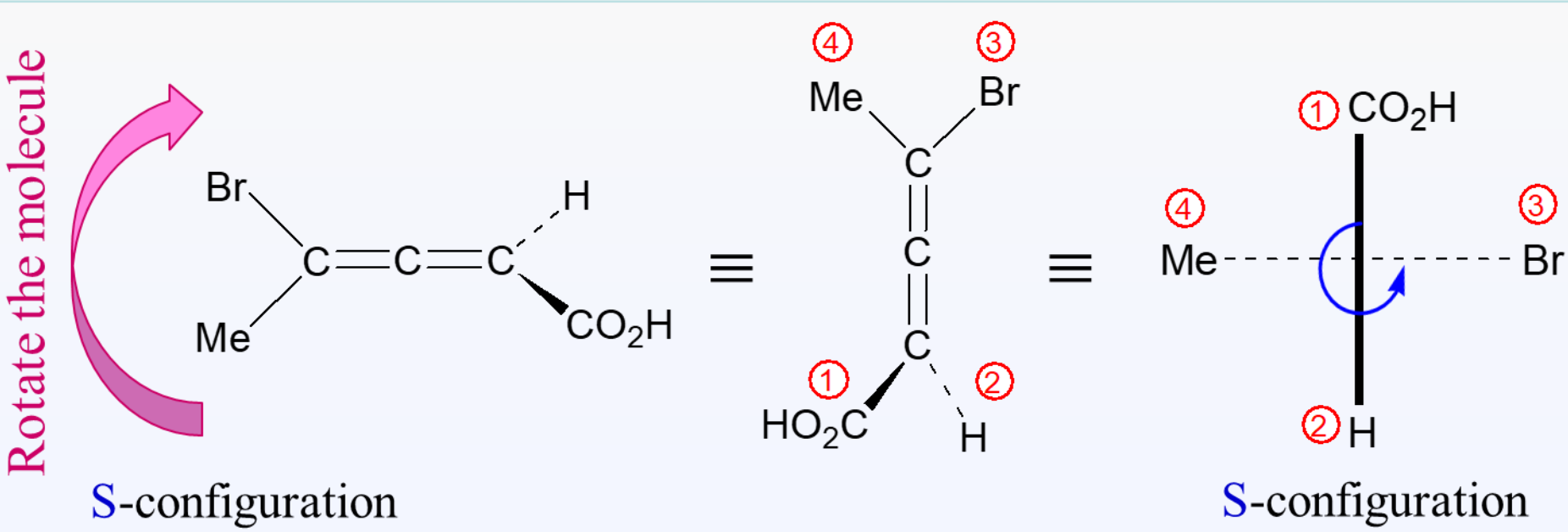


R-configuration

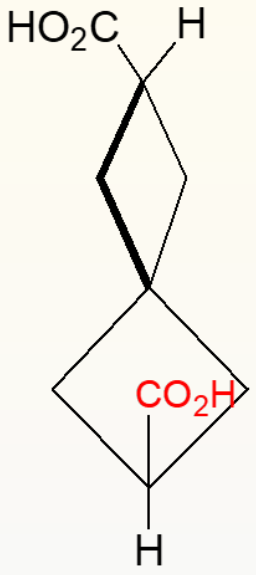
R-configuration

R/S Configuration (Allenes)

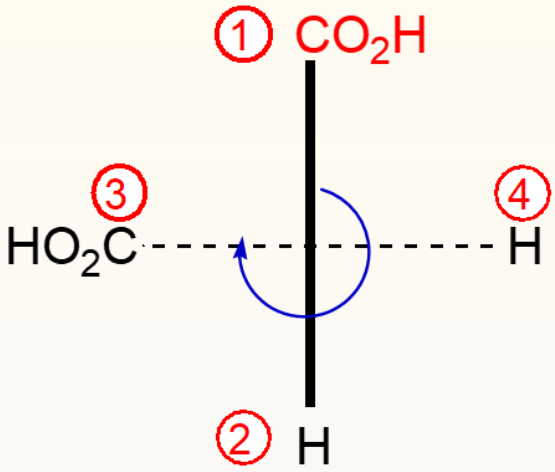




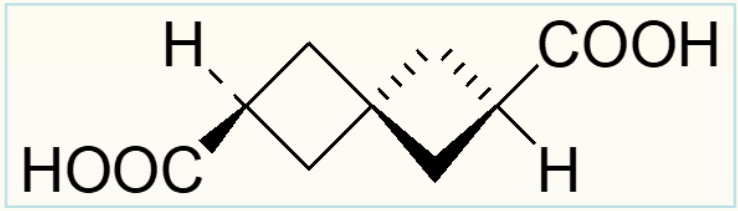
R/S Configuration (Spiranes)



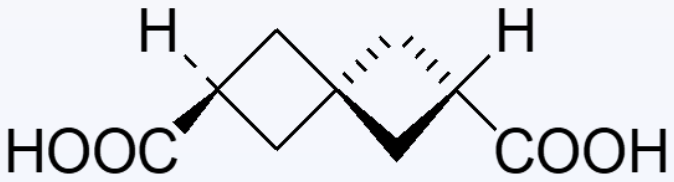
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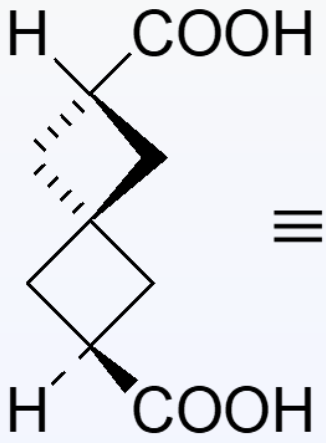
R-configuration



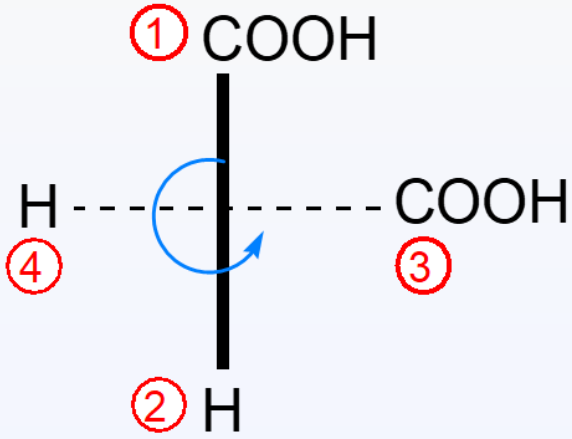
R-configuration



≡



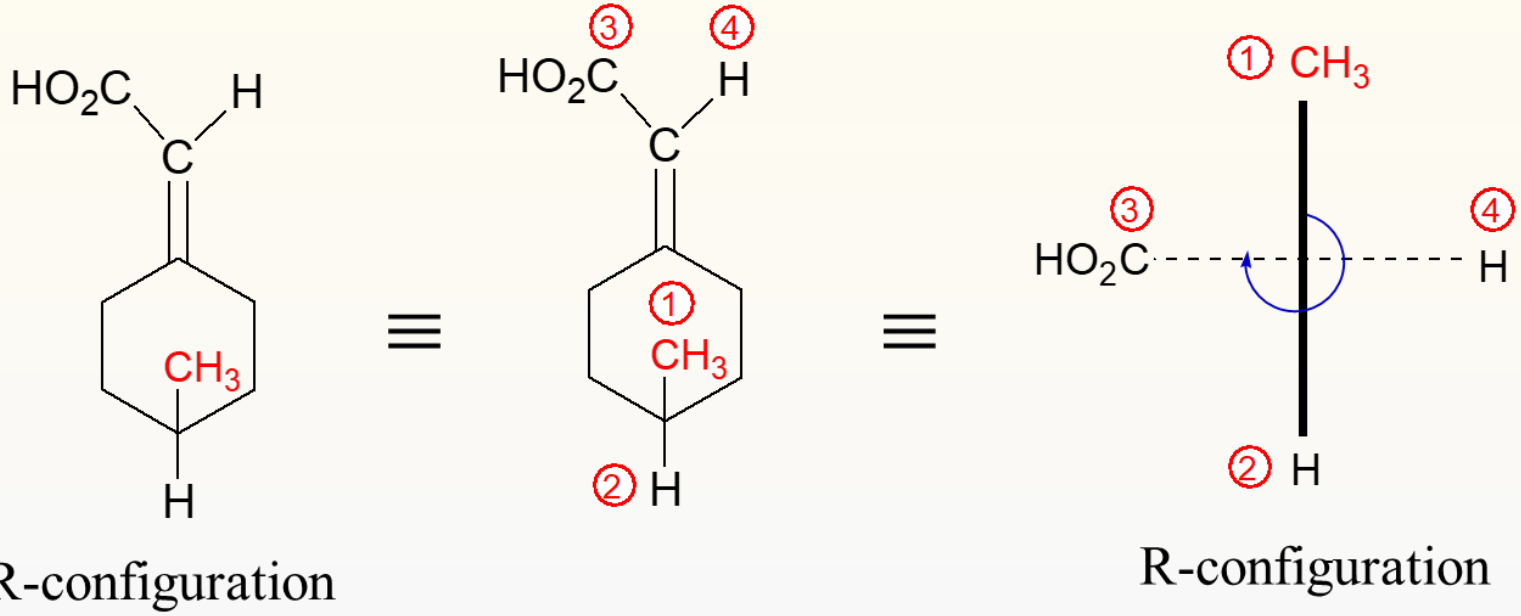
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S-configuration

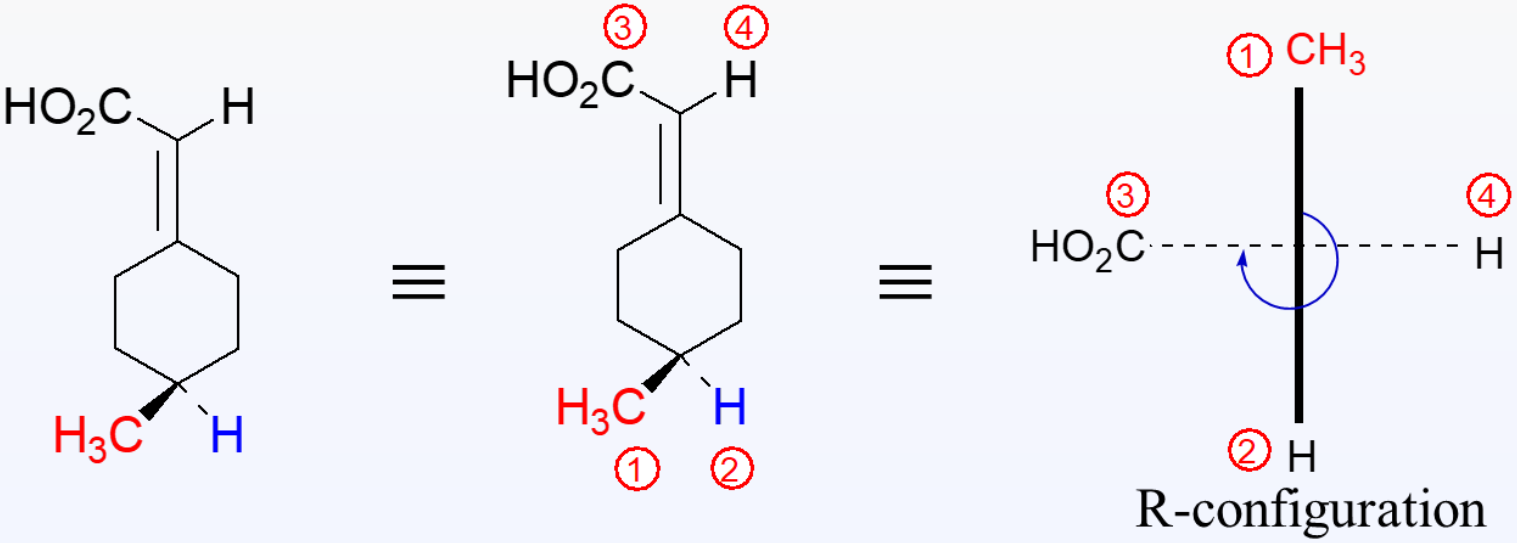
S-configuration

R/S Configuration (Alkylidene cycloalkanes)



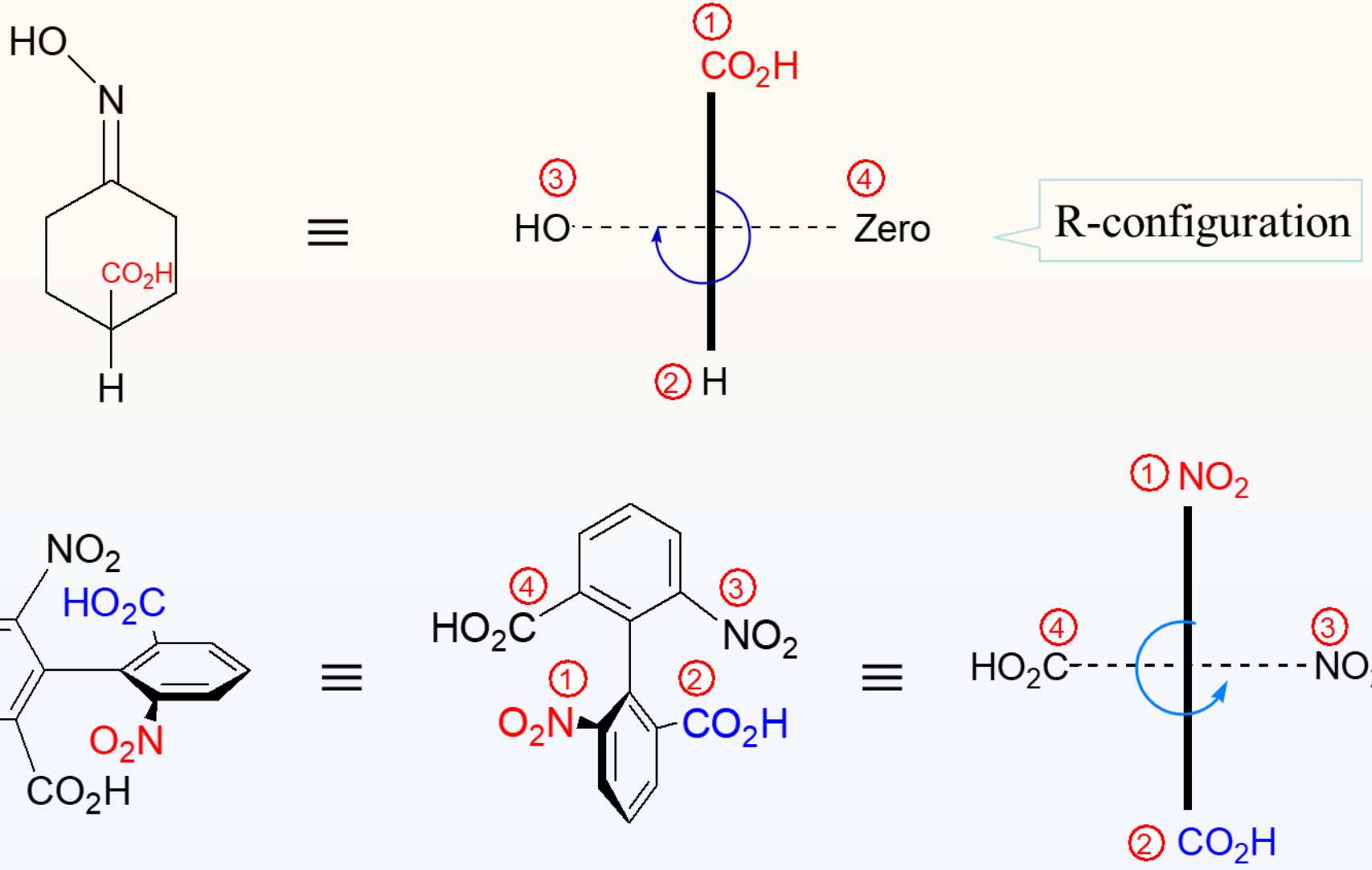
R-configuration

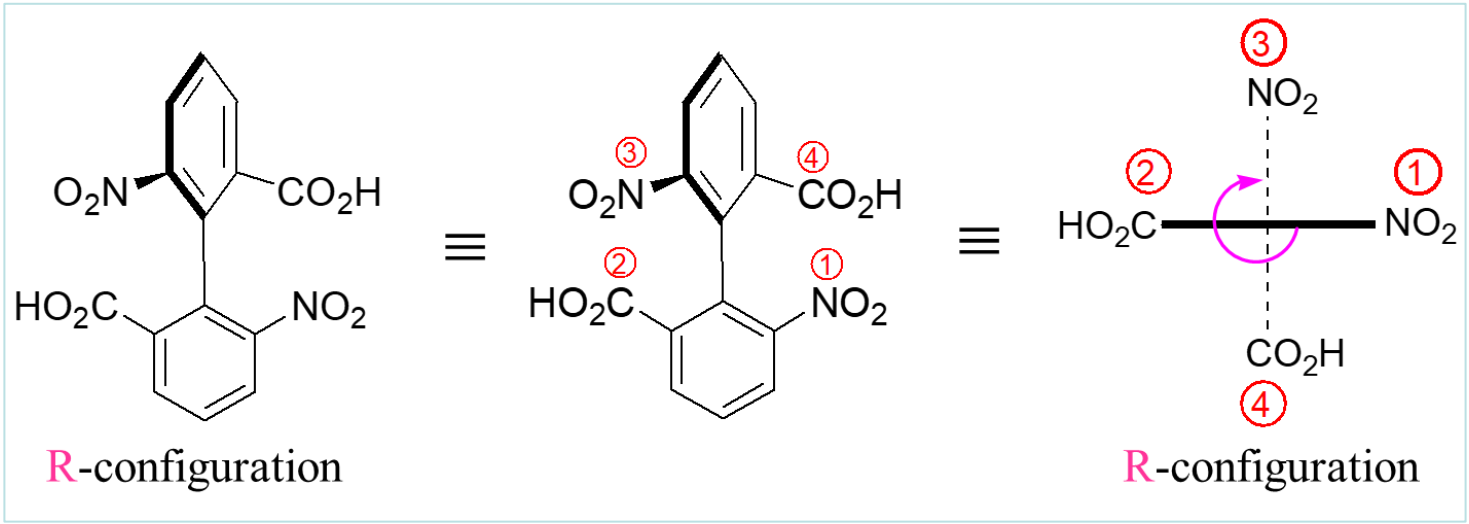
R-configuration



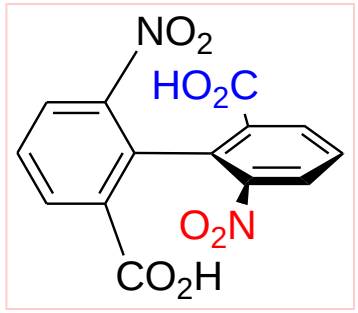
R-configuration

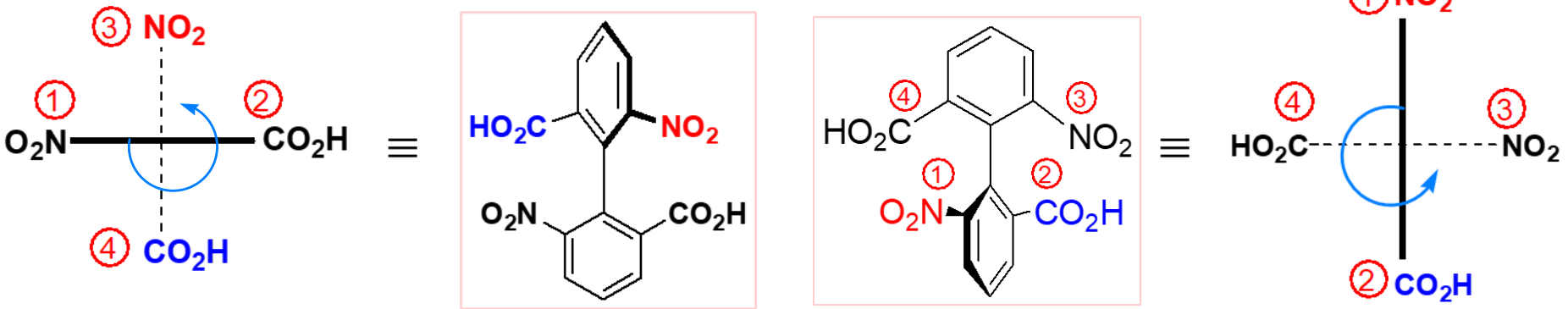
Near Groups Precede Far Groups



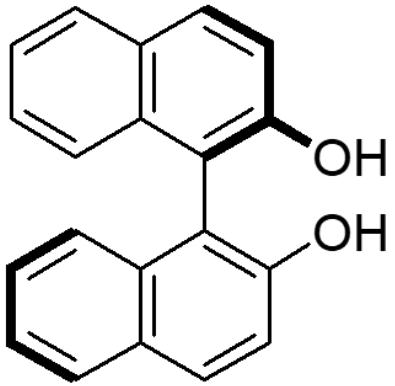


S-configuration

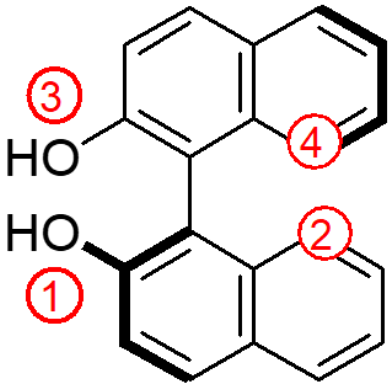




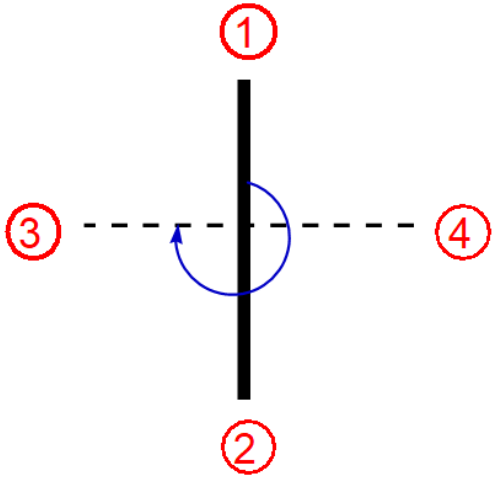
R/S Configuration



≡

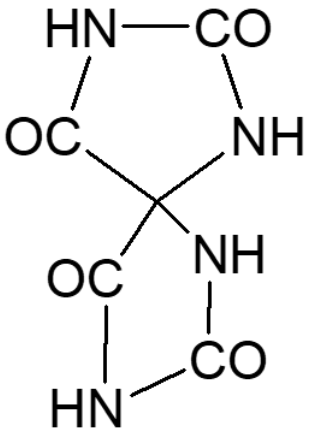


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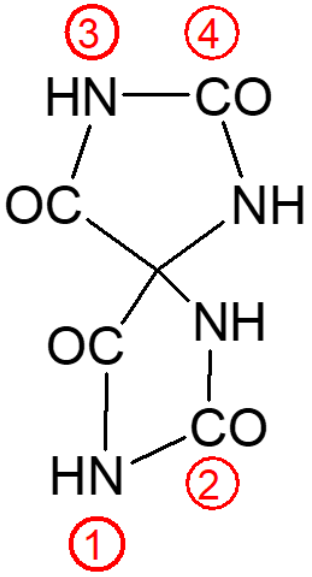


R-configuration

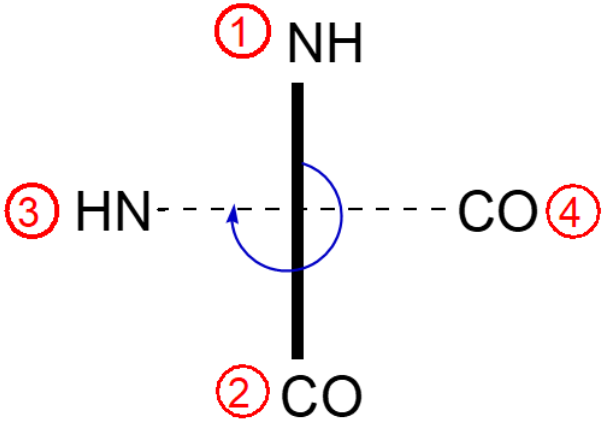
R-configuration



≡



≡

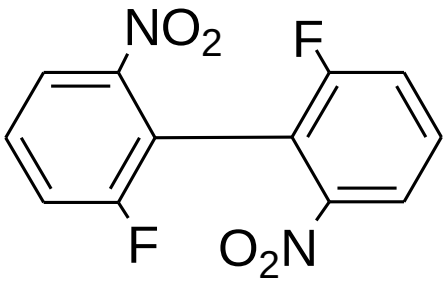


R-configuration

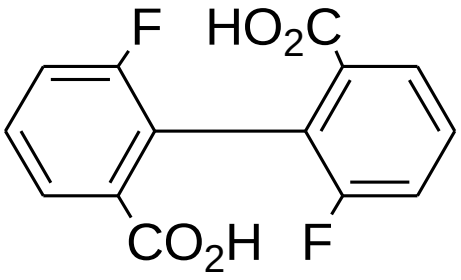
R-configuration

Practice Problems

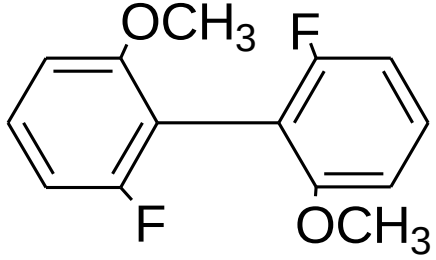
1. Comment on the resolvability, racemization and optical activity of the following biphenyl compounds:



(I)

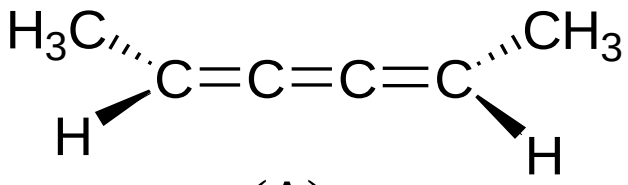


(II)

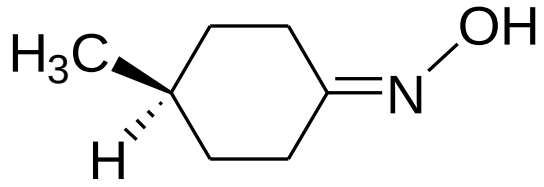


(III)

2. Which of the following compounds exhibit optical activity and which do not? Explain your reasoning and assign *R*/*S* configuration to the chiral compound.



(A)



(B)

Chem. 432 : Advanced Stereochemistry

✓ 1. Chiral and prochiral molecules, pro-R, pro-S, Homotopic and Heterotopic ligands and faces. Enantiotopic ligands (HCN addition) Diastereotopic ligands and faces. Enzymatic oxidation-reduction, prostereoisomerism in biochemical reactions (citric acid cycle), Stereochemistry of molecule synthetase reaction.

2. Configuration and conformation of cyclic compounds

3. Conformational analysis of biopolymers

⇨ 4. Chirality in molecules devoid of chiral centers: atropisomerism, biphenyls, allenes, cyclic allenes, molecules with planar chirality, cyclophanes, annulenes; chromatographic resolution, kinetic resolution, enzymatic resolution.

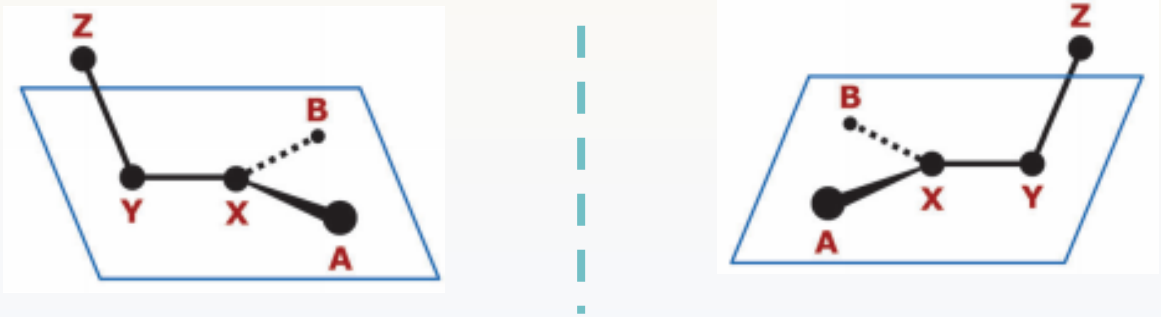
5. ORD curve, CD, octant rule, α-haloketone rule

6. Fused rings & bridged rings

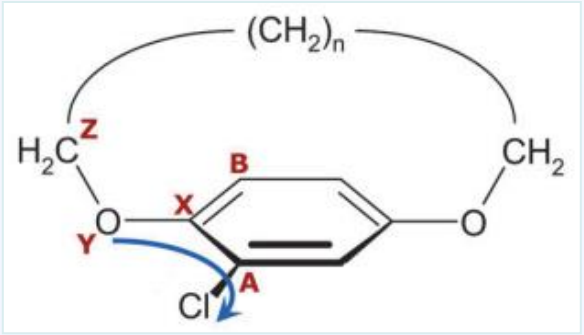
Chirality in Molecules Devoid of Chiral Centers

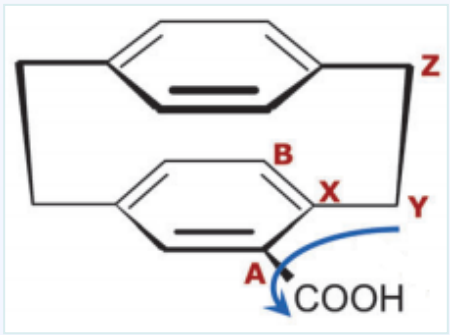
Plane of Chirality

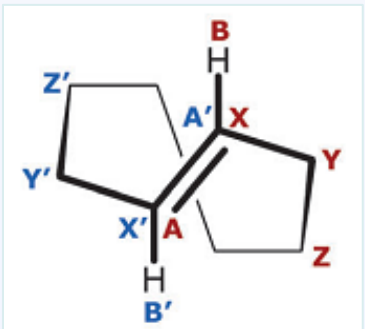
- The plane of chirality is a stereogenic element.
- The conditions for the existence of a plane of chirality are the presence of four coplanar atoms (labeled A, B, X, and Y in figure with A ≠ B), plus a fifth center Z located above (or below) the ABXY plane, and restricted in its rotation around the X–Y axis



Optically Active Molecules Due To Planar Chirality

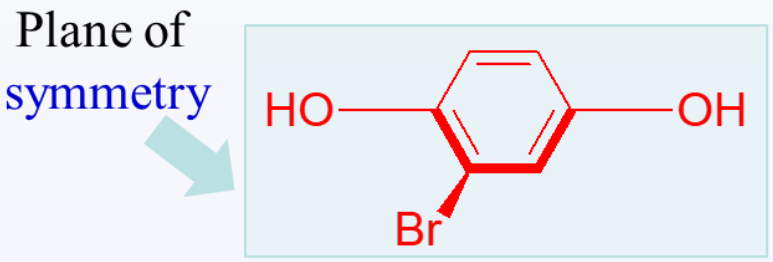






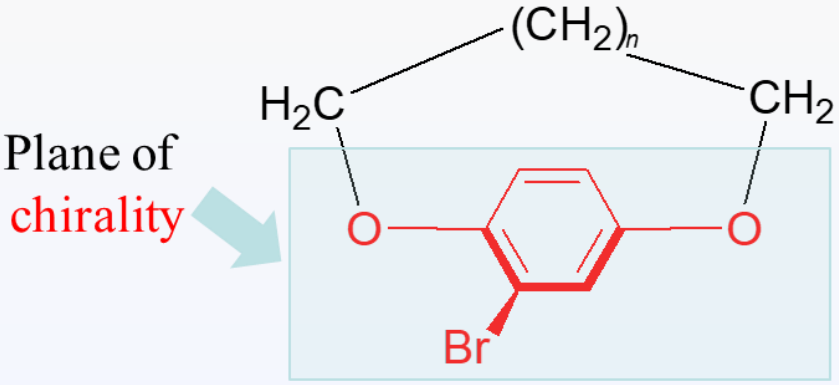
Chirality of *Ansa* Compounds

- The derivatives of hydroquinone are called *ansa* compounds.
- In *ansa* compounds, the two para positions of aromatic ring are attached to hetero atoms which in turn are connected through a polymethylene chain.
- In optically active *ansa* compounds, the benzene ring is dissymmetrically substituted and polymethylene bridge can be either above or below the plane of aromatic ring to give enantiomers.



Optically inactive

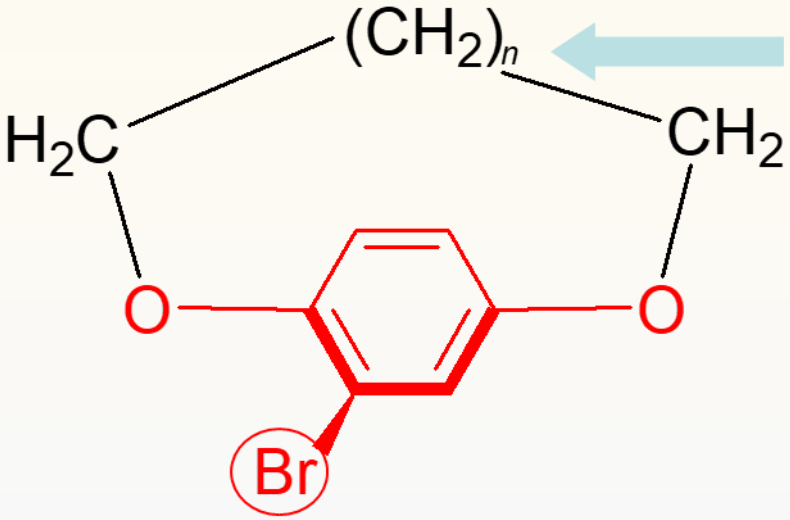
2-Bromohydroquinone



Optically active

Ansa compounds

Chirality of *Ansa* Compounds



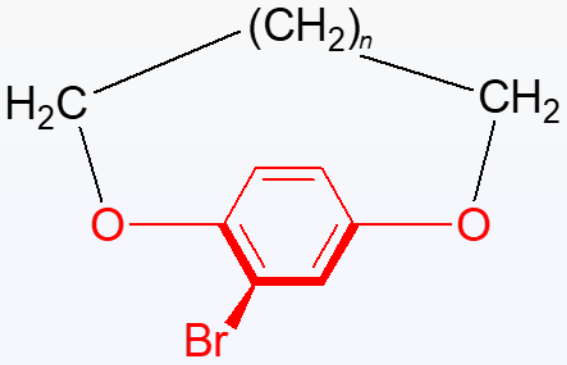
Ansa compounds

- When *n* = 8, optically active form of this *ansa* compound have been isolated.
- The methylene ring is perpendicular to the plane of the benzene ring.
- The –Br substituent prevents the rotation of benzene nucleus inside the large ring

- However, when *n* = 9, the racemization is very easy since now the out of plane movement of the benzene ring is no longer restricted due to steric factor and the two enantiomeric forms are not configurationally stable.

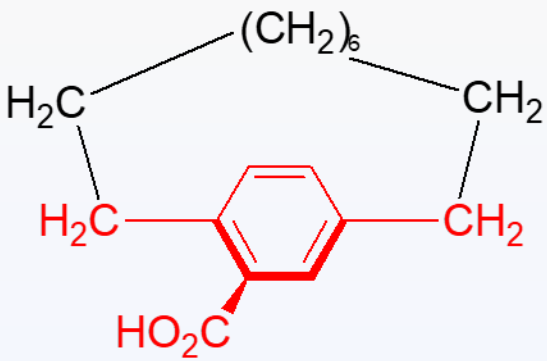
Chirality of Paracyclophanes

- A simple paracyclophane with one aromatic ring (compound II) which resembles an ansa compound (I) has been resolved.
- In other paracyclophanes (III) the two benzene rings are approximately parallel (arranged one above the other).
- To have enantiomerism, at least one of the rings must be dissymmetrically substituted.



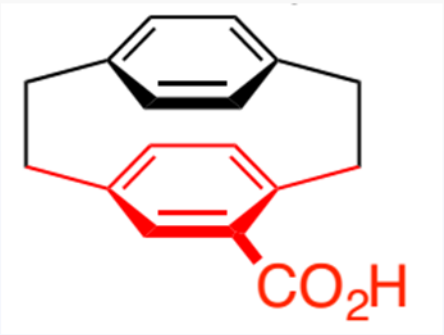
(I)

Ansa compounds



(II)

Paracyclophane

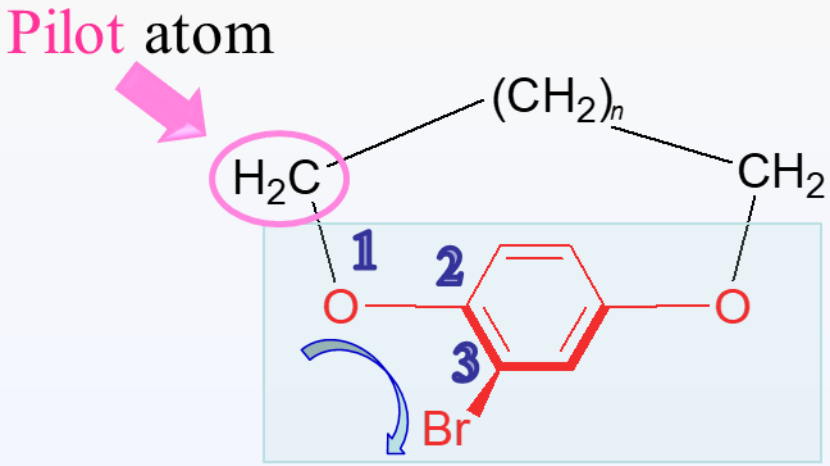


(III)

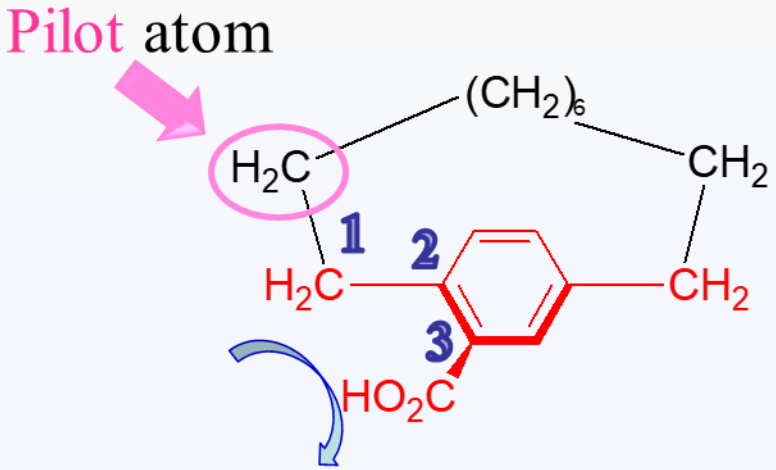
Paracyclophane

Configuration of Compounds with Planar Chirality

- To assign configuration to compounds with chiral planes the sequence rule is extended by choosing a pilot atom.
- Pilot atom is a priority atom directly linked to the plane but is not itself in the plane, thus it is first out of the plane atom.
- The pilot atom is chosen from that side of the plane which is most preferred.

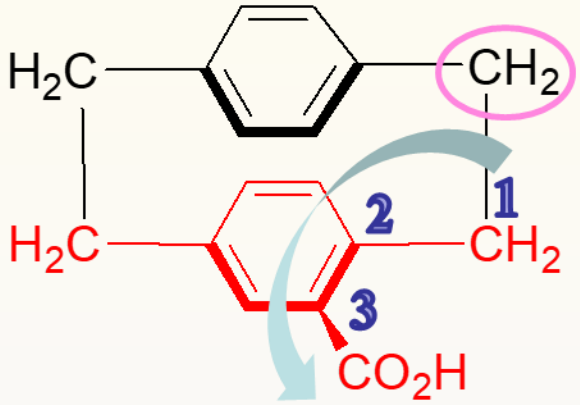


R- configuration

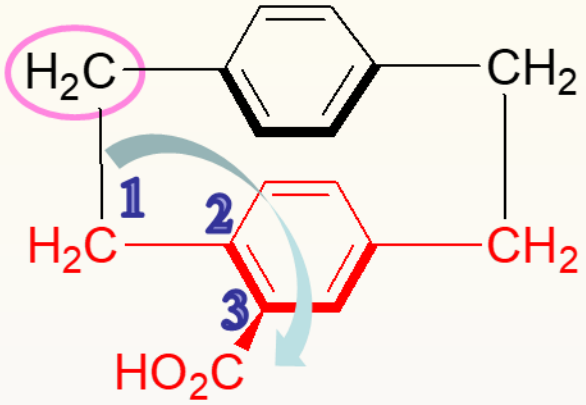


R- configuration

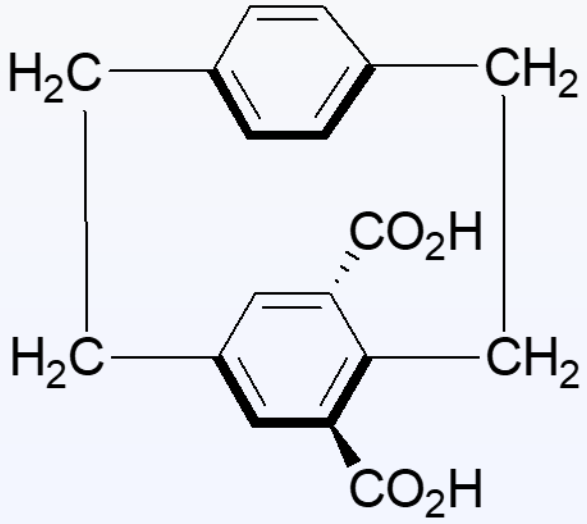
Configuration of Compounds with Planar Chirality



S- configuration



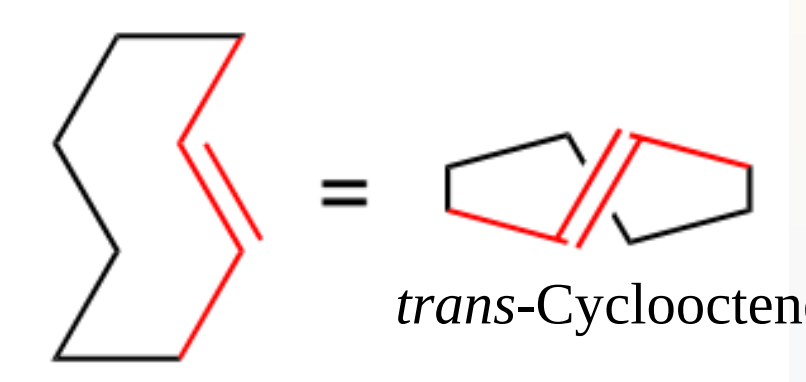
R- configuration

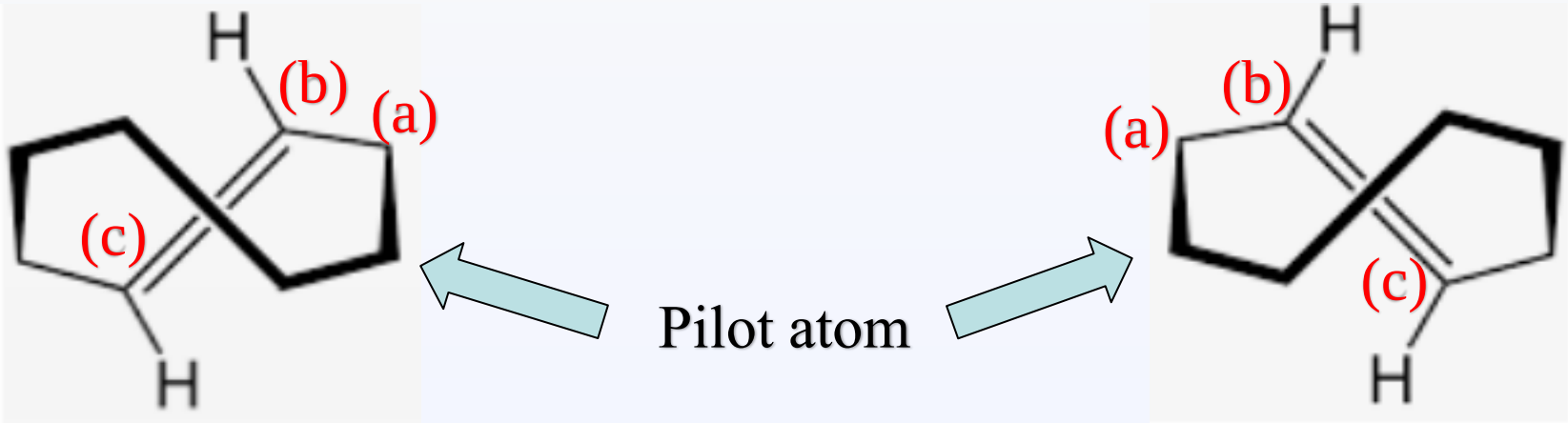


Optically inactive.
Plane of symmetry is present.
No plane of chirality.

Chirality of *trans*-Cyclooctene

- *trans*-Cyclooctene is chiral due to planar chirality: the two double bonded carbon atoms as well as the two hydrogens and the two carbon atoms directly adjacent constitute the chiral plane and polymethylene bridge is skewed in third dimension.

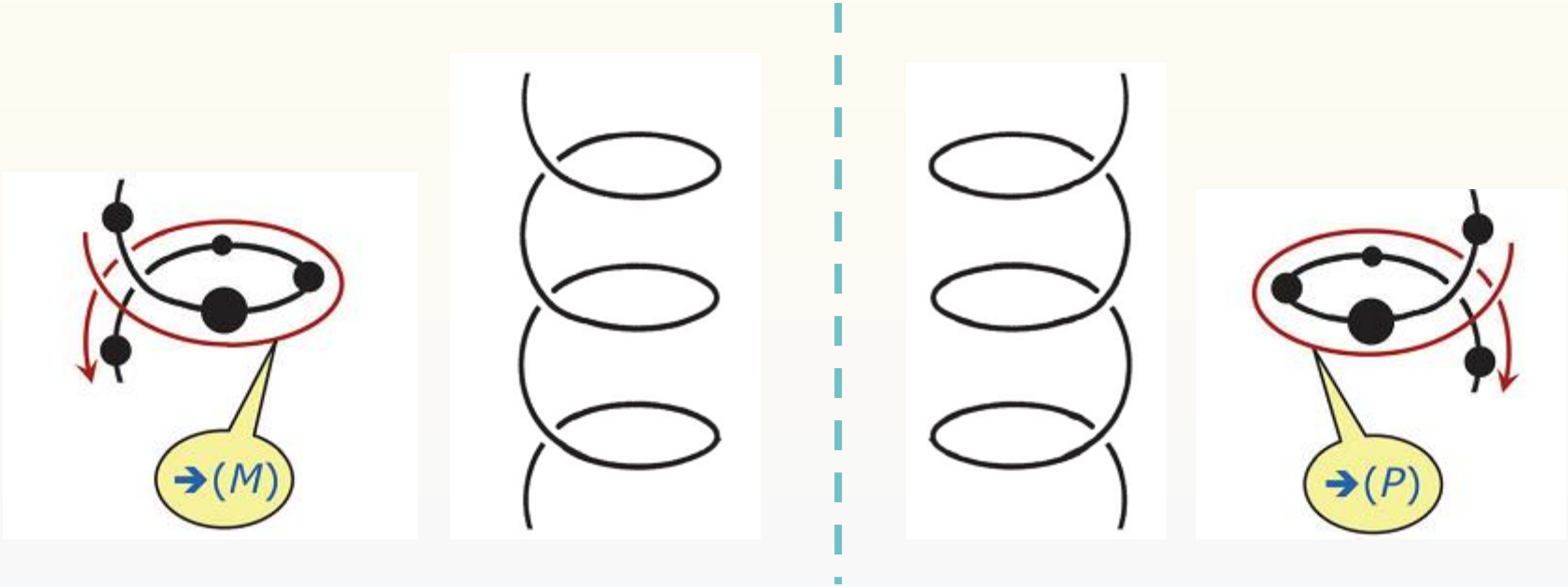




S- configuration

R- configuration

Optical Activity Compounds Due To Helicity

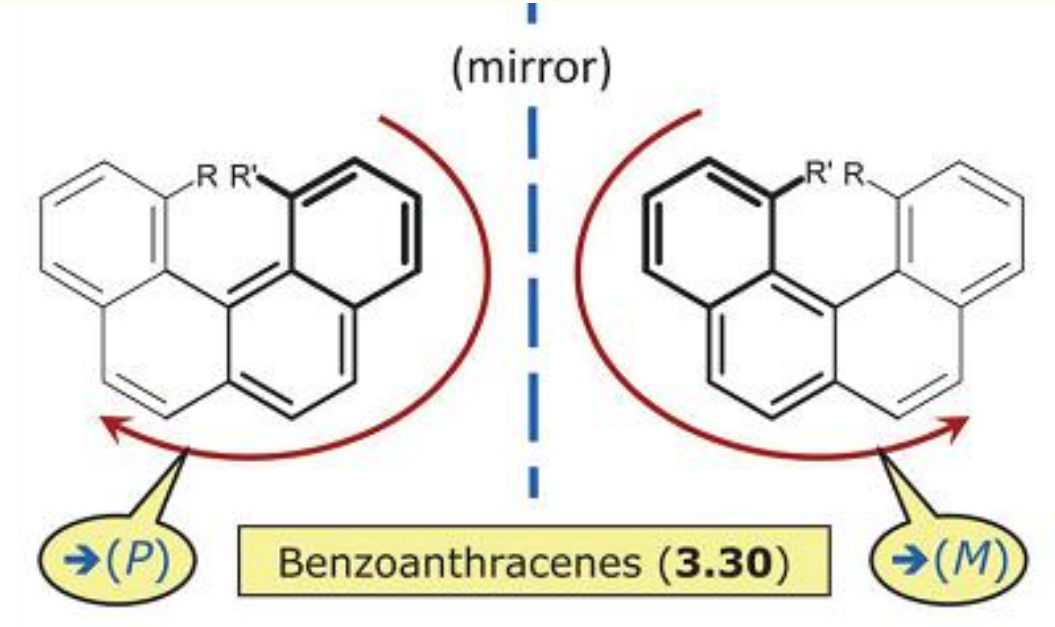


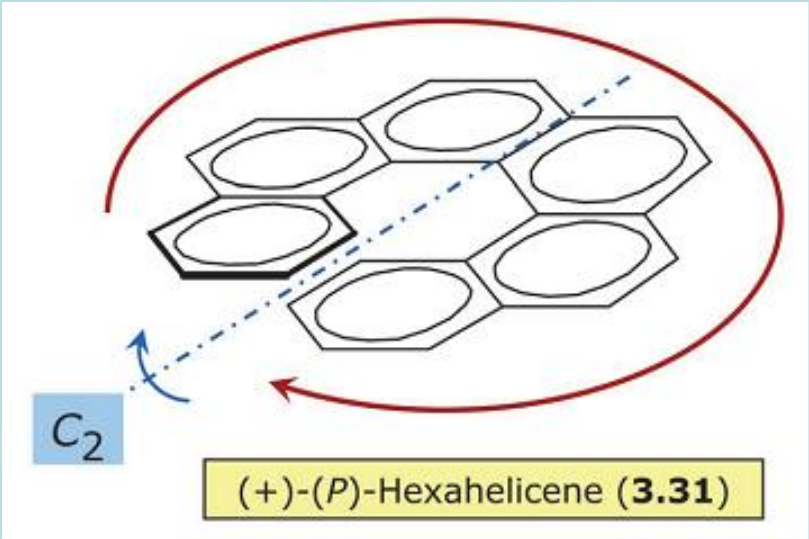
M- (minus, left handed)

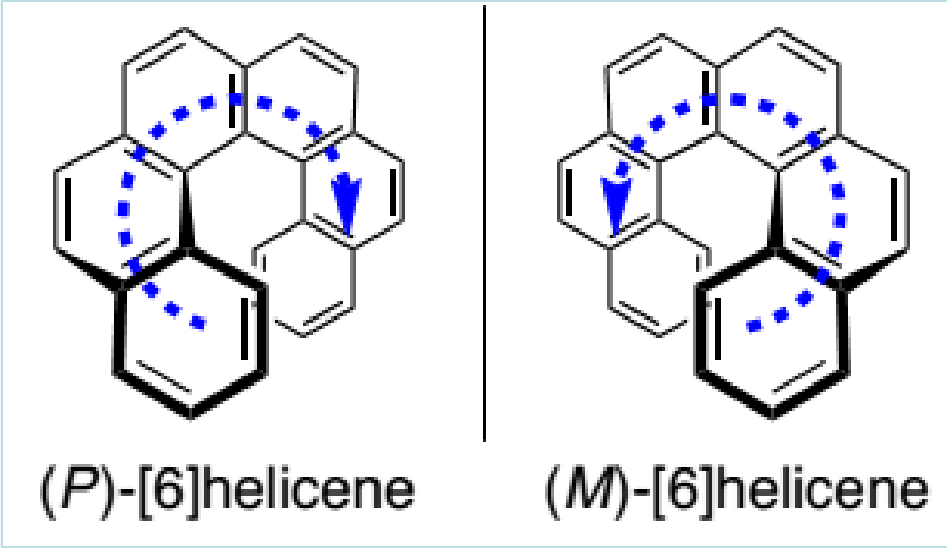
P- (plus, right handed)

- Helices are chiral objects : a right handed helix (a clockwise rotation) is designated P (plus) while a left-handed helix is *M* (minus).

Optical Activity Compounds Due To Helicity







Spirals in Our Life









